

MoCHII: A Three-Dimensional Monte Carlo Photoionization Code for Dusty Nebulae on Adaptive Octree Grids

KWANG-IL SEON ^{1,2}

¹*Korea Astronomy and Space Science Institute, 776 Daedeokdae-ro, Yuseong-gu, Daejeon 34055, Republic of Korea*

²*Department of Astronomy and Space Science, University of Science and Technology, Korea, 217 Gajeong-ro, Yuseong-gu, Daejeon 34113, Republic of Korea*

ABSTRACT

We present MoCHII (MOnTe Carlo for H II regions), a three-dimensional Monte Carlo radiative-transfer code for dusty photoionized nebulae on adaptive octree (AMR) grids. MoCHII is built by adding a self-consistent gas-physics layer to the validated octree transport, peel-off, and stochastic-dust modules of the dust code MoCAFE, and it computes hydrogen and helium photoionization and thermal balance, a trace-metal ionization cascade with modern analytic atomic data, the dust and PAH continuum, and the recombination-line and nebular-continuum diagnostics all from one radiation field. The band transport uses an analytic (zero-variance) direct-field estimator on the octree together with a dedicated Cartesian digital-differential-analyzer (DDA) backend, an explicit Monte Carlo diffuse field, dust absorption and forward scattering, and observer peel-off imaging. The atomic data are fitted offline and evaluated online: exact hydrogenic H I/He II cross sections, Verner et al. (1996) fits for the remaining ions, Badnell (2023) radiative and dielectronic recombination with a Mao & Kaastra (2016) ground-level split for case A/B, Voronov collisional ionization, and charge exchange, so that adding a metal ion is a data operation rather than a code operation. We validate the code against analytic attenuation (rate-integral bias +0.004%), the Strömgren sphere (ionized volume +0.56% versus an exact 1D reference; AMR identical to a uniform grid), a thermal H/He nebula (median 0.14% in T_e), and the Lexington 2000 H II region benchmarks, on which MoCHII lies within the published multi-code spread: the $n_p n_e$ -weighted temperature is 7894 K at HII40, inside the published 7720–8199 K, and 6376 K at HII20, 0.4% below the published floor; the outer radius and the He^+/H^+ ratio fall inside the published ranges at both, as do 22 of the 36 tabulated line entries, and $L(\text{H}\beta)$ is within 3% of them. The line-cooling data are cross-validated channel by channel against a CLOUDY c25.00 cooling decomposition on a fixed (T_e, n_e , ionization) state, where the summed line cooling agrees to 1%. Both benchmarks sit at the cool edge of the published range, which is the signature of the one documented limitation of the thermal loop: its cooling function is the zero-density limit, so the far-infrared fine-structure lines of low critical density ($[\text{O III}]$ 52 and 88 μm , $[\text{C II}]$ 158 μm) are not collisionally suppressed and overcool the line budget by 12% at $n_e = 110 \text{ cm}^{-3}$; a density-dependent $\Lambda(T, n_e)$ is the planned extension. Because the published columns are a quarter of a century old, we also run CLOUDY c25.00 on the same two models and carry it as an extra column: it is hotter than every published entry, MoCHII sits 4 and 8% below it, and several of the largest line-ratio differences (notably the $\text{S II}/\text{S III}$ split) prove to be drift in the published reference rather than in either modern code. We further demonstrate solution-driven ionization-front re-refinement (a re-refined octree reproduces a native high-resolution grid to +0.008% in ionized volume with $4.5\times$ fewer cells), the complete dust and PAH coupling (grain heating energy closure $L_{\text{IR}}/L_{\text{abs}} = 1.000$; the projected 8 μm image turning from a central peak into a limb-brightened ionization-front annulus as PAHs are destroyed in ionized gas), and an atomic photodissociation-region skin with photoelectric heating. MoCHII is written for modern three-dimensional synthetic observations of H II regions and their surroundings.

Keywords: H II regions (694) — Interstellar medium (847) — Radiative transfer (1335) — Photoionization (2060) — Interstellar dust (836) — Astronomy software (1855)

1. INTRODUCTION

Photoionized regions — H II regions, planetary nebulae, the diffuse ionized medium, and the circumgalactic gas illuminated by stars and active nuclei — are among the most information-rich objects in astrophysics. Their emission-line and continuum spectra encode the ionizing radiation field, the elemental abundances, the electron temperature and density, and the dust content of the gas. Extracting that information requires a radiative-transfer model that couples the ionizing radiation field to the ionization, thermal, and dust state of the gas self-consistently.

The classic tools for this problem are one-dimensional: CLOUDY (G. J. Ferland et al. 2017; M. Chatzikos et al. 2023) and MAPPINGS (R. S. Sutherland & M. A. Dopita 1993, 2017) solve the coupled ionization, thermal, and (in the case of MAPPINGS) shock structure of a spherical or plane-parallel slab with very detailed microphysics: large ion inventories, density-dependent multilevel cooling, non-equilibrium and time-dependent options, and grain and PAH physics. Their strength is completeness of the local physics; their limitation is geometry. Real H II regions are clumpy, aspherical, and shadowed, and the diffuse recombination field they produce is intrinsically three-dimensional.

Three-dimensional Monte Carlo photoionization codes were developed to address exactly this geometry. MO-CASSIN (B. Ercolano et al. 2003, 2005, 2008) pioneered the self-consistent Monte Carlo treatment of the direct and diffuse fields on a Cartesian grid, with a large ion and dust inventory, and it defines the *de facto* three-dimensional benchmark suite (the Lexington/Meudon comparison cases; G. Ferland 1995; D. Péquignot et al. 2001). The code of K. Wood et al. (2004) demonstrated Monte Carlo photoionization with an efficient path-length estimator for the mean intensity. M³ (Messenger Monte Carlo MAPPINGS V; Y. Jin et al. 2022a,b) grafts three-dimensional Monte Carlo transport onto the extensive MAPPINGS V microphysics, combining broad atomic physics with arbitrary Cartesian geometry. CMACIONIZE (B. Vandenbroucke & K. Wood 2018) adds a task-based, unstructured (Voronoi or adaptive Cartesian) grid and a modern software design. These codes share a common architectural constraint: their spatial grids are either uniform Cartesian or, in the CMACIONIZE case, unstructured, and their diffuse-field and grid-traversal designs are tightly coupled.

MoCHII (MONte Carlo for H II regions)³ is a new three-dimensional Monte Carlo photoionization code built around a different design choice: it reuses the mature octree adaptive-mesh-refinement (AMR) engine, event-driven traversal, peel-off imaging, and stochastic dust/PAH emission of the dust radiative-transfer code MoCAFE (Seon, in preparation)⁴, and adds a gas-physics layer on top. The motivation is practical. Retrofitting an octree traversal onto an existing uniform-grid photoionization code is difficult: the ray-segment bookkeeping is entangled with the physics loop, and structural rewrites are error-prone (the MO-CASSIN `pathSegment` traversal, for instance, carries a dual position representation with many mutation sites). An event-driven AMR transport engine, by contrast, already *is* the design such a retrofit would have to reach. MoCAFE provides that engine, validated for dust transport, so MoCHII inherits octree AMR, an analytic direct-field estimator, a dedicated Cartesian digital-differential-analyzer (DDA) backend, peel-off synthetic imaging, MPI-3 shared-memory parallelism, and HDF5/FITS I/O without modifying the dust tree.

On top of that engine MoCHII adds: (i) an ionizing frequency band with threshold-aligned bins and a zero-variance direct-field estimator; (ii) hydrogen and helium photoionization and thermal balance with modern recombination data; (iii) a trace-metal ionization cascade organized as a data-driven registry, so that adding an ion (currently eleven elements, C through Ca) is an offline data operation; (iv) an explicit Monte Carlo diffuse field with a case-A/B option; (v) the complete dust and PAH coupling in the ionizing and far-ultraviolet (FUV) bands, including equilibrium and stochastic dust emission; (vi) solution-driven ionization-front re-refinement of the octree; and (vii) recombination-line, collisional-line, and nebular-continuum diagnostics with leaf-resolved emissivity output for synthetic imaging. The atomic data are fitted offline from CHIANTI and the Strathclyde/Badnell tables and evaluated by analytic formulae at run time, with a provenance header on every coefficient file.

The remainder of this paper is organized as follows. Section 2 compares MoCHII with the established 1D

³ <https://github.com/seoncafe/MoCHII>

⁴ <https://github.com/seoncafe/MoCafe>

and 3D codes. Section 3 describes the Monte Carlo transport, the grids, the diffuse field, the iteration and convergence scheme, and the quasi-random launch option. Section 4 documents the atomic data and the offline fitting pipeline. Section 5 presents the validation against analytic solutions and the Lexington benchmarks. Section 6 demonstrates the solution-driven I-front re-refinement and a cosmological post-processing application. Section 7 presents the dust and PAH coupling, and Section 8 the photodissociation-region skin. Section 9 summarizes and outlines planned extensions.

2. COMPARISON WITH EXISTING CODES

Table 1 places MoCHII next to the reference codes along the axes most relevant to three-dimensional nebular modeling. The comparison is deliberately architectural rather than a claim of superior physics: CLOUDY and MAPPINGS carry more complete local microphysics (larger ion inventories, density-dependent cooling, non-equilibrium and shock options), and MOCASSIN carries a comparably broad ion and dust inventory in three dimensions. MoCHII’s distinguishing features are the octree AMR grid with solution-driven re-refinement, the reuse of a validated dust-transport engine (so that dust absorption, forward scattering, equilibrium and stochastic emission, and PAH physics are first-class rather than bolted on), and the peel-off synthetic-imaging output.

The grid axis is the sharpest distinction. MOCASSIN and M³ use uniform Cartesian grids; CMAcIONIZE uses an unstructured Voronoi or adaptive Cartesian grid; MoCHII uses an octree whose refinement can be driven by the ionization solution itself (Section 6). On the Strömgren problem this lets MoCHII concentrate resolution at the ionization front and reproduce a native high-resolution run with several times fewer cells.

The diffuse-field axis separates the on-the-spot (OTS) approximation from explicit transport. MoCHII defaults to case-B OTS (no diffuse packets, consistent with the analytic direct-field estimator) but offers an explicit Monte Carlo diffuse field with case-A rates and ground-recombination emission, matching the MOCASSIN and CLOUDY treatment; the benchmark runs of Section 5.2 use it, so that the transfer treatment is the same on both sides of that comparison.

The atomic-data axis reflects vintage. MoCHII uses exact hydrogenic cross sections, Verner et al. (1996) fits, Badnell (2023) recombination, CHIANTI v11 collision strengths, and Mao & Kaastra (2016) level-resolved radiative recombination — the same modern data that CLOUDY uses — rather than the older compilations still carried by MOCASSIN.

The metal-treatment axis is the clearest algorithmic difference. MOCASSIN, M³, CMAcIONIZE, and the 1D codes solve a *full network*: the ionization of every element is one coupled nonlinear system in which the metal opacity consumes photons from the radiation field, the metal electrons enter n_e , and the metal photoheating enters the thermal balance, all iterated to self-consistency (radiation field $\leftrightarrow$ all ions $\leftrightarrow n_e \leftrightarrow T$). MoCHII instead adopts a *trace-species* treatment: because the metals are present at abundances $\sim 10^{-4}$ – 10^{-3} , they are separated as a perturbation on the H/He (and dust) skeleton — the radiation field and n_e are set by H, He, and dust, and each element’s ionization cascade is an independent, effectively linear chain evaluated on the converged (J_ν, n_e, T) . The one feedback that is *not* dropped is metal line cooling, which enters the thermal balance because the H II-region temperature is itself largely the result of metal cooling (Section 4.4). Where the trace separation genuinely breaks down, it is restored selectively by switches (the default is trace, with recoupling only where the physics demands it): `ion_metal_abs` adds the metal photoionization opacity to the band (on by default), `metal_ne` feeds the metal electrons into the n_e closure — essential in neutral photodissociation-region gas, where carbon dominates the electron budget, and which the calculation of Section 8 uses — and `metal_heat` adds metal photoheating to the thermal balance. The design payoff is that, with each element an independent linear chain, adding an ion is a data-file addition rather than a change to a coupled solver (the registry of Section 4), and the cell solve stays at the level of closed-form evaluations rather than a coupled-network iteration.

A related but separate axis is the cooling inventory — which channels a code carries at all, since that is what sets the equilibrium temperature. As implemented in the distributed source, the Monte Carlo code of K. Wood et al. (2004) cools by ten five-level metal ions (N I, N II, O I, O II, O III, Ne III, S II, S III, C II, C III) plus two two-level ions ([N III] $57\ \mu\text{m}$ and [Ne II] $12.8\ \mu\text{m}$), and CMAcIONIZE by the same ten five-level ions plus three two-level ions, adding [S IV] $10.5\ \mu\text{m}$; both then add free-free emission and hydrogen and helium recombination cooling, in each case with the H^+ and He^+ terms only. MoCHII carries line cooling for 28 ions — H I and the 27 metal ions of the eleven-element registry (Appendix A) — and adds three channels that neither of those two codes carries: H I collisional excitation, collisional-ionization cooling of H and He, and the He^{++} terms of the free-free and recombination sums. In a $10^4\ \text{K}$ H II region these are small (H I collisional lines are 0.7% of the HII40 cooling budget), but they are the

terms that take over in the hot, partly neutral gas of an ionization front and in the shielded gas beyond it.

3. METHODS

MOCHII follows the conventions of its parent dust code: all input is a single Fortran namelist (`par`); computation is in double precision; parallelism is MPI-only with MPI-3 shared-memory windows for the grid; output is HDF5 with a FITS fallback; and algorithm variants are selected by procedure pointers at startup. A run reads (or builds) a grid, persists a gas leaf state, and iterates transport and equilibrium to convergence.

3.1. The ionizing band and the source

The ionizing band spans photon energies $[E_{\min}, E_{\max}] = [13.598, 100]$ eV in n_ν bins (typically 32). The source spectrum — a Planck function $B_\nu(T_\star)$ or a tabulated spectrum in shape or physical (per-eV, per-Hz, per-Å, per- μm) units — is integrated on a sub-grid within each bin and normalized so that $\sum_b L_b = L_{\text{band}}$. Packets sample the bin from the luminosity CDF and carry $L_{\text{pkt}} = L_{\text{band}}/N$.

Threshold-aligned bins.—By default (`ion_align_edges`) the bin *edges* are pinned onto the ionization thresholds — H I (13.6 eV), He I (24.6 eV), He II (54.4 eV), and the active metal edges gathered from the registry — so that no bin straddles a threshold; the sub-segments between thresholds are log-uniform. This removes a real discretization error: a coarse 32-bin log grid straddles the 24.6 eV He I edge and delivers $\sim 18\%$ too many He-ionizing photons against a true 40 kK blackbody ($Q_{\text{HeI}}/Q_{\text{H}} = 0.127$ versus the exact 0.108), over-ionizing helium and, downstream, the metals. Aligning the edges restores the exact ratio at fixed bin count; this is the single largest lever on the He and metal ionization structure in the HII40 benchmark (Section 5.2). When thresholds outnumber bins the lowest-priority metal edges are dropped gracefully, so the option is safe at any bin count, and disabling it recovers the plain logarithmic grid bit-for-bit.

FUV extension and composite sources.—With `par%add_fuv` the band grows a second, lower segment of FUV bins on $[E_{\text{FUV}}, E_{\min}]$ (default $E_{\text{FUV}} = 6$ eV) below the unchanged ionizing bins. H and He cross sections vanish below their thresholds, so the FUV bins interact only with dust and with the low-threshold metal stages, producing the neutral-zone He I-like species (e.g. Mg II, C II) that the ionizing-only band cannot. The band is fed by any number of internal point sources (0–16) plus, independently, one isotropic external radiation field. The external field is a band-integrated mean intensity J entering through the box faces or a bounding

sphere; the inward flux is the hemisphere integral of $J \cos \theta$, so the entering band power is $L = \pi J A_{\text{surf}}$, and entry directions are cosine-weighted (Lambert), reproducing an isotropic interior field of energy density $u = 4\pi J/c$. Three analytic interstellar-radiation-field presets (Draine, Habing, Mathis) are available for the FUV segment.

3.2. Opacity and the zero-variance edge walk

Each leaf carries the band absorption coefficient

$$\kappa_b = n_{\text{H}} [x_{\text{HI}} \sigma_{\text{HI}}(E_b) + y_{\text{He}} (x_{\text{HeI}} \sigma_{\text{HeI}} + x_{\text{HeII}} \sigma_{\text{HeII}})(E_b)] d_{\text{cm}} (+ \kappa_{\text{dust}}), \quad (1)$$

stored as the full (n_ν, n_{leaf}) array, plus optional metal photoionization opacity. Because the ionizing band has no scattering (without dust) and the diffuse field, when used, is emitted explicitly, each packet’s contribution to the mean-intensity tally is computed *analytically* along its ray — the forced-first-flight estimator of L. B. Lucy (1999), made exact:

$$\Delta J_b(\text{leaf}) = w L_{\text{pkt}} \frac{e^{-\tau_{\text{in}}} - e^{-\tau_{\text{out}}}}{\kappa_b}, \quad (2)$$

accumulated leaf by leaf until the domain edge. No interaction sampling is needed; the only Monte Carlo variance is in the direction and bin sampling. From the reduced tally J_b the code forms, for each absorber i , the photoionization rate and photoheating rate

$$\Gamma_i = \sum_b \frac{[4\pi J \Delta\nu]_b \sigma_i(E_b)}{h\nu_b},$$

$$\mathcal{H}_i = \sum_b [4\pi J \Delta\nu]_b \sigma_i(E_b) \left(1 - \frac{E_{\text{th},i}}{E_b}\right). \quad (3)$$

3.3. Grids: octree AMR and the Cartesian DDA backend

MOCHII inherits the MOCaFE octree. The grid may be read as an AMR leaf list, built from a namelist size $(n_x \times n_y \times n_z)$ with an optional uniform-density sphere or shell, or read from a 3D density cube. Geometry is fully analytic behind accessor functions, so a leaf’s center and half-size are array reads for a general octree and raster-index decodes for a uniform Cartesian (“car”) grid — the same floating-point expressions in both cases. The car backend allocates no tree or neighbor arrays at all (a saving that scales to ~ 10 GB at 512^3) and traverses rays with a dedicated incremental Amanatides–Woo DDA walk, which is the default for uniform grids. A verification mode reproduces the single-level octree traversal bit-for-bit, and the DDA agrees with it to rounding, at up to twice the transport throughput. The two backends share all of the physics, transport, and peel-off code.

Table 1. Architectural comparison of MoCHII with reference photoionization codes. “OTS” = on-the-spot (case B); “explicit” = transported diffuse field. This table compares design, not the breadth of local microphysics, on which the 1D codes and MOCASSIN are more complete.

	MoCHII	MOCASSIN	M ³	CMAcIonize	Cloudy / MAPPINGS
Dimensionality	3D	3D	3D	3D	1D
Grid	octree AMR + Cartesian DDA	uniform Cartesian	uniform Cartesian	Voronoi / adaptive Cartesian	spherical / plane-parallel
Solution-driven refinement	yes (I-front)	no	no	static AMR	1D shells
Diffuse field	OTS default; explicit MC option	explicit MC	explicit MC	explicit MC	outward-only / OTS
Recombination data	Badnell 2023 + Mao & Kaastra 2016	Verner & Ferland 1996	MAPPINGS V	Verner / Badnell	Badnell / internal
Metal treatment	trace-species registry (data-driven)	full network	full MAPPINGS network	full network	full network
Dust / PAH	absorption + forward scattering + equilibrium + stochastic + PAH survival	absorption + emission	MAPPINGS grains	absorption + emission	detailed grains/PAH
Parallelism	MPI-3 shared memory	MPI	MPI	task-based / MPI	OpenMP / serial
Synthetic output	peel-off images, line maps, IR SED	line maps, images	line maps	line maps, images	1D spectra

3.4. Ionization and thermal equilibrium

In each leaf, the ionization state is solved (`solve_ion_cell`) from the ionization rates $R_i = \Gamma_i + n_e C_i(T)$ and the case-A/B recombination rates α_i : $x_{\text{HI}} = \alpha n_e / (R_{\text{HI}} + \alpha n_e)$, the He chain $n_{i+1}/n_i = R_i / (\alpha_{i+1} n_e)$, closed by $n_e = n_{\text{H}}[x_{\text{HII}} + y_{\text{He}}(x_{\text{HeII}} + 2x_{\text{HeIII}})]$ and solved by a damped fixed point on n_e . When thermal balance is enabled, the temperature is updated once each iteration by finding the root of the net rate $\mathcal{N}(T) = \mathcal{H}(T) - \mathcal{C}(T)$ by bisection on $\log T_e$; the ionization state is re-solved at every trial T . The cooling $\mathcal{C}(T)$ assembles recombination cooling, free-free (with the net ion charge Z , including the trace metals at $Z_{\text{ion}} = \text{stage} - 1$), collisional ionization, H I line cooling, and the trace-metal cooling sum $\sum n_e n_{\text{ion}} \Lambda_{\text{ion}}(T)$. Optional secondary ionization by fast photoelectrons (J. M. Shull & M. E. van Steenberg 1985) redistributes the photoelectron energy between heating and secondary H I/He I ionization in partially neutral gas.

3.5. The iteration loop and convergence

Each iteration zeroes the band tally, transports the stellar packets (and, if enabled, rebuilds and transports the diffuse packets), reduces the tally over MPI, forms the rate integrals, solves each leaf, and refills the opacity

from the new state; the stellar tally is recomputed from zero each iteration. Every rank runs the identical cell solves from the reduced tallies, so all ranks agree on convergence, and only the node master writes the shared state.

Convergence is measured two ways every iteration, and `par%conv_crit` selects which pair controls the loop: the largest change found in any single cell ($\max |\Delta x_{\text{HII}}|$, $\max |\Delta T_e|/T_e$) and the volume-integrated measures

$$\frac{\sum_{\text{leaf}} V |\Delta x_{\text{HII}}|}{\sum_{\text{leaf}} V x_{\text{HII}}}, \quad \frac{\sum_{\text{leaf}} V n_e |\Delta T_e|}{\sum_{\text{leaf}} V n_e T_e}. \quad (4)$$

The volume measures fix two stall modes of these single-cell maxima: ionization-front jitter occupies little volume (the front shell is a few thousand cells against $\sim 10^5$ ionized ones, so its Monte Carlo noise enters the L_1 norm suppressed by that ratio), and the n_e weight removes the vacuum and no-heating cells whose floor-pinned oscillation dominates $\max |\Delta T_e|$ in FUV/PDR configurations. On a FUV dusty Strömgren case at 8×10^6 packets, where these single-cell maxima stall at 0.1 and 6.4, the volume measures decay smoothly to Monte Carlo floors of 2.7×10^{-3} and 6.6×10^{-3} by iteration ~ 27 , and 30 further iterations leave the ionized volume unchanged to 0.014%. The floors scale as $1/\sqrt{N_{\text{packets}}}$, and the volume criterion is the recommended production setting.

3.6. Explicit diffuse field

When the explicit diffuse field is enabled, case-B OTS is replaced by case-A rates plus explicit emission of ground-recombination packets in three channels (H II, He II, He III) with rate $\alpha_1 = \alpha_A - \alpha_B$ and photon energies $E = E_{\text{th}} + kT_e x$, $x \sim \text{Exp}(1)$. The emission table is rebuilt from the current state each iteration; packets are transported by the same edge walk. The sampled photon energy is mapped to its bin by a binary search of the stored bin-edge array, correct on both the aligned and the plain logarithmic grids: an energy exactly on a threshold edge falls in the bin *above* it, the physically correct side for a recombination photon emitted just above threshold. An optional fourth channel adds the He I excited-level recombination radiation with the low-density branching of D. E. Osterbrock & G. J. Ferland (2006). At HII40 the diffuse band luminosity converges to $\sim 0.5 L_*$.

3.7. Quasi-random photon launching

Optionally the packet launch draws its variables — source component, frequency bin, direction, and any entry-surface coordinates — from an Owen-scrambled Sobol sequence (S. Joe & F. Y. Kuo 2008; B. Burley 2020) instead of the Mersenne Twister, while every post-launch transport event stays pseudo-random. The Sobol point is addressed by the *global* photon number, so the launch set is independent of the MPI task count. Because the ionizing band has no scattering, the direct field is an analytic function of the launch variables (Eq. 2), so a low-discrepancy launch set reduces its noise directly: on the fixed-state analytic-attenuation test the cell-wise Γ_{HI} error scales as $\sim 1/N$ (Monte Carlo $1/\sqrt{N}$), with effective photon gains of $29\times$ at $N = 2^{17}$ and $222\times$ at 2^{20} , at unchanged wall time and no significant bias.

3.8. Peel-off imaging

After convergence, one extra transport pass runs with observer peeling (the MOCAFE convention adapted to the band). Every emitted packet peels a direct contribution $L_{\text{pkt}} w e^{-\tau}/(4\pi r^2)$ toward each observer, and, when dust scattering is sampled, every interaction peels a scattered contribution with the Henyey–Greenstein phase function. The optical depth to the box edge is integrated by a tally-free walk, so the images carry exactly the transport’s extinction. Images are TAN projections written as surface brightness, split into direct/scattered and ionizing/FUV channels; with a spectral cube option the channel axis becomes the band bin for hardness maps. Leaf-resolved line and continuum emissivities are also written, from which flux-conserving line and field maps are produced.

4. ATOMIC DATA

The design principle for the atomic data is that adding an ion is a *data operation*, not a *code operation*. All rate coefficients are fitted offline by a Python pipeline from CHIANTI v11.0.2 (K. P. Dere et al. 1997; R. P. Dufresne et al. 2024) and the Strathclyde/Badnell tables, and are evaluated online by fixed analytic formulae. Every coefficient file carries a provenance header (source, version, fit range, maximum fit error), so the coefficients are auditable and the Fortran only evaluates formulae.

4.1. Photoionization cross sections

The one-electron ions H I ($Z = 1$) and He II ($Z = 2$) use the exact ground-state hydrogenic cross section (D. E. Osterbrock & G. J. Ferland 2006)

$$\varepsilon = \sqrt{E/E_{\text{th}} - 1},$$

$$\sigma(E) = \frac{\sigma_0}{Z^2} \left(\frac{E_{\text{th}}}{E} \right)^4 \frac{\exp(4 - 4 \arctan \varepsilon/\varepsilon)}{1 - e^{-2\pi/\varepsilon}}, \quad (5)$$

with $\sigma_0 = 6.30 \times 10^{-18} \text{ cm}^2$ and E_{th} the true ionization potential (13.598 and 54.416 eV), tied to the physical edge so it stays aligned with the band grid. He I and every metal ion use the single-shell analytic fit of D. A. Verner et al. (1996):

$$x = \frac{E}{E_0} - y_0, \quad z = \sqrt{x^2 + y_1^2},$$

$$\sigma(E) = \sigma_0 [(x-1)^2 + y_w^2] z^{-Q} \left(1 + \sqrt{z/y_a} \right)^{-P},$$

$$Q = 5.5 - \frac{1}{2}P, \quad (6)$$

with parameters parsed directly from the authors’ `phfit2.f` table. For the elements absent from that table (P, Cl, K — Opacity-Project gaps), the D. A. Verner & D. G. Yakovlev (1995) outer-shell subshell fit is used instead, the same hybrid as MOCASSIN. The cross sections are cross-checked against the independent implementation in the exoplanet-atmosphere code EXHALE.

4.2. Recombination

The default recombination model takes the total (case-A) radiative recombination α_A from the N. R. Badnell (2006) radiative recombination fit,

$$\alpha_{\text{RR}}(T) = \frac{A}{\sqrt{T/T_0} (1 + \sqrt{T/T_0})^{1-b} (1 + \sqrt{T/T_1})^{1+b}},$$

$$b = B + C e^{-T_2/T}, \quad (7)$$

the ground-level ($n = 1$) direct-capture rate α_1 from the level-resolved parameterization of J. Mao & J. Kaastra (2016), and forms case B by removing the ground

term, $\alpha_B = \alpha_A - \alpha_1$. For the H-like ions this α_1 is derived from the exact hydrogenic cross sections (Eq. 5) under detailed balance, so it is exact; the $He\text{ II} \rightarrow He\text{ I}$ channel adds the Badnell three-term dielectronic sum. This construction mirrors how CLOUDY splits case A and case B (total minus resolved ground) and reuses the same modern data. An alternative setting selects the L. Hui & N. Y. Gnedin (1997) fits; the two agree to $<0.6\%$ for H II and He III, the one substantive difference being $He\text{ II} \rightarrow He\text{ I}$, where the Badnell value ($4.37 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$ at 10^4 K) is $\sim 7\%$ higher and coincides with CLOUDY, while the older L. Hui & N. Y. Gnedin (1997) value is low and the MOCASSIN Verner & Ferland value (4.84) is the high outlier. For the metal cascade, total recombination is the N. R. Badnell (2006) radiative recombination plus the Badnell-group dielectronic fits $\alpha_{\text{DR}}(T) = T^{-3/2} \sum_i c_i e^{-E_i/T}$ (N. R. Badnell et al. 2003), read from the Strathclyde tables (verified md5-identical to the copies CLOUDY redistributes), with a CHIANTI fallback for the few ions Badnell does not tabulate.

4.3. Collisional ionization and charge exchange

Collisional ionization uses the G. S. Voronov (1997) fit

$$k(T) = A \frac{1 + P\sqrt{U}}{X + U} U^K e^{-U}, \quad U = \frac{\Delta E}{kT}, \quad (8)$$

with an optional hybrid that multiplies each rate by the constant K. P. Dere (2007)/Voronov ratio per (element, stage); the difference matters only in shielded, low-ionization zones where collisions are the sole metal-ionization channel. Charge exchange with hydrogen uses C. Huang et al. (2023) Table 4 for the first stages and the J. B. Kingdon & G. J. Ferland (1996) fits $k = a T_4^b (1 + c e^{dT_4})$ for the higher stages (recombination direction). The tabulated higher-stage rates are indexed by the *product* ion of the reaction $X^{i+} + H^0 \rightarrow X^{(i-1)+}$, a convention worth stating explicitly because it is easy to misread by one ionization stage and because the doubly and triply ionized metal fractions are sensitive to it; the indexing used here is checked against MOCASSIN's ionization loop and CMAIONIZE.

4.4. Cooling and n -level diagnostic data

Collisional line cooling enters the code at two levels, both built from the same CHIANTI v11.0.2 model of each ion (level energies from the observed term structure, radiative A -values, and effective collision strengths descaled after A. Burgess & J. A. Tully 1992). The thermal loop needs a cooling rate at every trial temperature of the bisection, and therefore uses a closed-form fit; the

diagnostic output needs density-dependent emissivities, and therefore solves the level populations.

The thermal-loop coefficient is a multi-exponential fit

$$\Lambda_{\text{ion}}(T) = T^{-1/2} \sum_{i=1}^N A_i e^{-T_i/T} \quad [\text{erg cm}^3 \text{ s}^{-1}], \quad (9)$$

a rate coefficient multiplying $n_e n_{\text{ion}}$, fitted over 10^3 – 10^5 K to the $n_e \rightarrow 0$ limit of that ion's own statistical-equilibrium solution — all population in the lowest level, excitation at the observed level energies, and the full cascade energy radiated. The maximum fit error over the temperatures that carry the cooling is 2.2% across the default C/N/O/Ne/S set (the coefficients of every ion are distributed with the code, Appendix A; the heavily depleted Si/Cl/Ca ions have larger full-range fit errors, confined to the temperature tails where the cooling contribution is negligible).

The zero-density limit is a deliberate approximation with a stated domain of validity: $\Lambda_{\text{ion}}(T)$ carries no collisional de-excitation, so it is exact for lines whose critical density is far above the local n_e and increasingly too large for those below it. The affected lines are the far-infrared fine-structure transitions, [O III] 52 and $88 \mu\text{m}$ and [C II] $158 \mu\text{m}$, whose electron critical densities are 10^2 – $10^{3.5} \text{ cm}^{-3}$, so at the $n_e \sim 10^2 \text{ cm}^{-3}$ of an H II region the fit overestimates the line-cooling budget by about 12% (Section 5.3), while it is accurate in the diffuse gas the limit is written for. A density-dependent $\Lambda(T, n_e)$ table, which the same pipeline can produce and which the planetary-nebula densities of Section 9 require in any case, is the planned replacement.

For the output-time diagnostics, the lowest n levels, radiative A -values, and effective collision strengths $\Upsilon(T)$ (low-order Chebyshev fits in Burgess–Tully scaled space; worst transition 1.7% , Table 5) are exported for the line-emitting ions of the eleven-element registry. A statistical-equilibrium solve at output time (`nlevel.mod`) uses these to compute the collisional emission-line luminosities exactly — the density-dependent emissivities the low-density cooling fits cannot provide. The fitted files reproduce direct CHIANTI spline evaluations of the classic diagnostic ratios to 0.05 – 0.5% , and the Fortran solver matches an independent Python reference solver to four significant digits. PyNeb cross-checks give [O III] 5007/4363 to 1.2% and [N II] 6584/5755 to 0.3% ; the density diagnostics and the iron lines differ by up to 6 – 16% from PyNeb's independent collision data — the well-known database spread, a data-vintage bracket rather than a code error.

4.5. Recombination lines and the nebular continuum

Hydrogen and He II recombination-line emissivities are interpolated from the P. J. Storey & D. G. Hummer (1995) case-A/B tables (following `par%case_ab`), and He I lines from the R. L. Porter et al. (2012, 2013) case-B collisional-radiative emissivities. The nebular continuum combines the free-bound plus free-free coefficients of the MOCASSIN (B. Ercolano & P. J. Storey 2006) tables with a Milne extension above the edges and the three two-photon continua; the free-free Gaunt factor is the D. G. Hummer (1988) fit, with the P. A. M. van Hoof et al. (2014) table as a drop-in option (the two agree to a median 5×10^{-5} in T_e). A He I 2^3S metastable diagnostic, including the 10830 Å optical depth, is also available.

5. VALIDATION

MoCHII is validated against analytic solutions, an exact 1D reference solver, and the Lexington benchmark suite, in staged tests. Each AMR result is required to reproduce the equivalent uniform-grid run.

5.1. Rate integrals and the Strömgren sphere

The zero-variance edge walk (Eq. 2) is verified on a uniform sphere ($n_H = 1$, $R = 1$ pc, $x_{HI} = 0.1$) with a central 10^5 K source. Against the cell-volume-averaged analytic attenuation, Γ_{HI} has a mean bias of +0.004% and a median deviation of 0.38% (pure Monte Carlo noise), and Γ_{HeI} a bias of +0.011% (Figure 1).

The Strömgren test uses a uniform $n_H = 100$ sphere, $Q_H = 10^{49} \text{ s}^{-1}$ ($T_\star = 4 \times 10^4$ K), case B, and $T_e = 10^4$ K. For the reference we solve the spherically symmetric ionization-equilibrium equations directly, with a deterministic radial marching over 2×10^4 shells that carries the same band bins, photoionization cross sections, and recombination and collisional rates as the code: each bin’s photon flux is attenuated with the running optical depth evaluated at mid-shell, and the H/He ionization equilibrium is solved shell by shell. The sweeps must update opacities Gauss–Seidel style (a Jacobi sweep, using the previous sweep’s opacities, advances the front only one absorption length in each pass); a few sweeps then converge. “Exact” means free of Monte Carlo noise and grid-converged at this resolution — in the sharp-edge limit it reproduces the analytic Strömgren radius $R_S = [3Q_H/(4\pi n_H^2 \alpha_B)]^{1/3} = 3.153$ pc, from which the full solution differs physically (helium consumes photons; the front has a finite width). The 1D reference gives an effective ionized radius $R_{\text{eff}} = (3V_{\text{ion}}/4\pi)^{1/3} = 3.0365$ pc; a uniform level-6 octree gives 3.0534 pc (+0.56%), and a front-refined octree with $3.5\times$ fewer cells reproduces it to four decimals, while a center-refined control (coars-

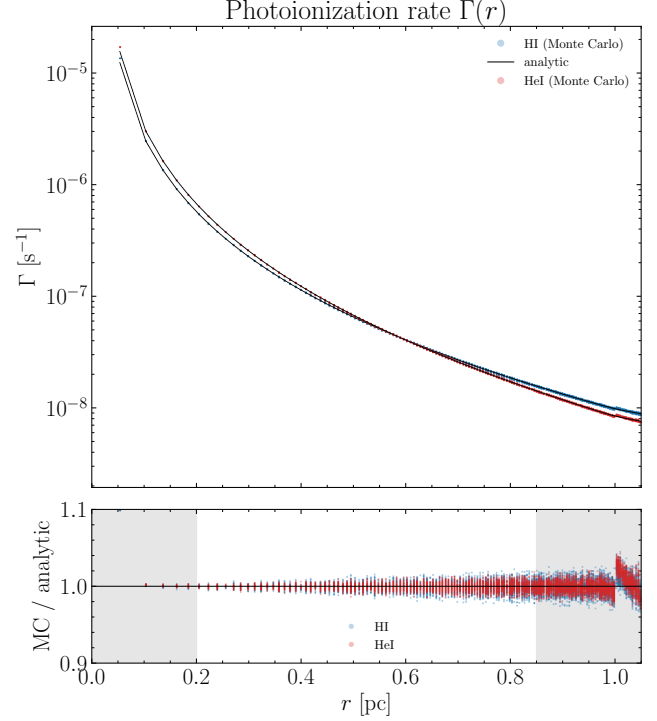


Figure 1. Rate-integral test: $\Gamma(r)$ from the Monte Carlo tally (points) against the analytic point-source attenuation (lines); the lower panel shows the ratio. Shaded regions are excluded (cell-size effects at the center, the voxelized sphere edge outside).

est cells at the front) deviates by +2.35% — the ionization front is the one place needing resolution (Figure 2). With the thermal balance on, $T_e(r)$ matches the 1D thermal reference to a median 0.14% and a mean bias +0.006% (Figure 3); without metals the metal-free nebula runs at 19–24 kK, the known overshoot that motivates the trace-metal cooling.

5.2. The Lexington benchmarks

The HII20 and HII40 cases of the Lexington/Meudon comparison (G. Ferland 1995; D. Péquignot et al. 2001) use 20 and 40 kK blackbody sources ($Q_H = 1.0$ and $4.3 \times 10^{49} \text{ s}^{-1}$), $n_H = 100$ in a hollow shell, and the standard Lexington abundances (He 0.1, C 2.2×10^{-4} , N 4×10^{-5} , O 3.3×10^{-4} , Ne 5×10^{-5} , S 9×10^{-6}). They are the natural reference for a new code because eight independent codes published results for the same two models, so the benchmark carries its own measure of how large a disagreement is significant. Tables 2 and 3 place MoCHII next to those eight, in the compilation of B. Ercolano et al. (2003), together with the Monte Carlo code of K. Wood et al. (2004) that was added to the same tables afterwards.

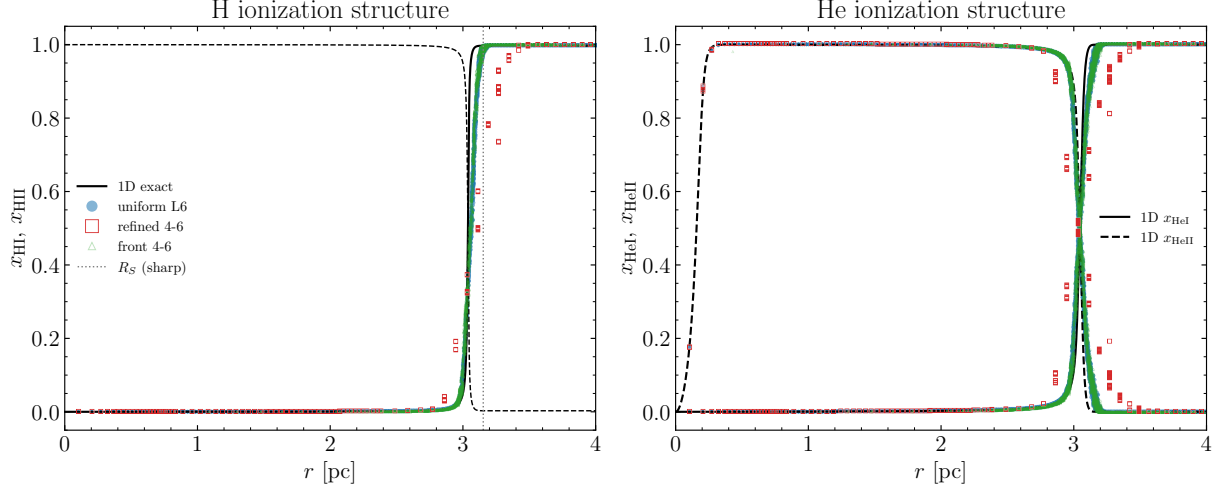


Figure 2. Strömgren test: H and He ionization structure against the 1D exact reference (black); uniform level-6 (blue), center-refined (red), and front-refined (green, overlying blue).

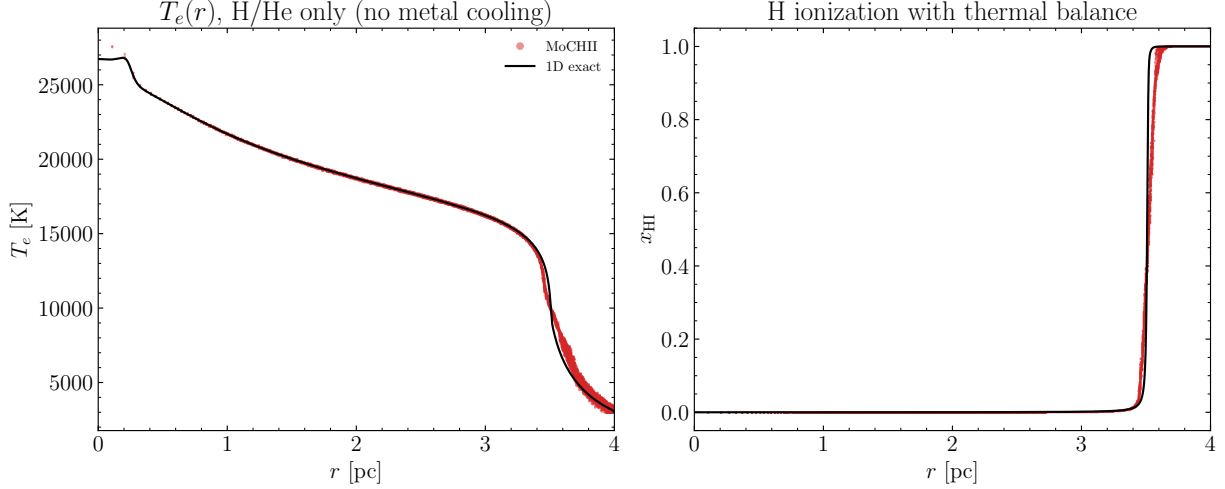


Figure 3. Thermal test: $T_e(r)$ and $x_{\text{HI}}(r)$ with the thermal balance, against the 1D exact reference.

618 The MoCHII runs compared here are configured to
 619 match the published calculations wherever the setup is
 620 a choice rather than a physical difference. The ionizing
 621 luminosity is set from the benchmark blackbody lumi-
 622 nosity so that Q_{H} reproduces the specified 4.265×10^{49}
 623 (HII40) and $1.004 \times 10^{49} \text{ s}^{-1}$ (HII20); the band carries
 624 32 threshold-aligned bins (Section 3.1); and the diffuse
 625 field is transported explicitly with case-A rates and the
 626 He I channels (Section 3.6), the treatment the published
 627 codes use, so that the comparison is not confounded by
 628 the transfer treatment. The recombination-line emissiv-
 629 ities stay on the case-B tables, because the Lyman lines
 630 are optically thick in these models whatever the contin-
 631 uum treatment is; the two cases are separate switches
 632 (`par%case_ab` for the ionization, `par%line_case` for the
 633 line tables). The grid is a uniform level-6 octree (64^3

634 leaves) run with 4×10^7 stellar packets to the volume-
 635 integrated convergence criterion.

636 Those published columns are, however, a quarter of a
 637 century old, and one of them (GF) is an early version
 638 of a code that is still developed. To separate what has
 639 changed in MoCHII from what has changed in the refer-
 640 ence itself, we also ran CLOUDY c25.00 (C. M. Gunasek-
 641 era et al. 2025) on both benchmarks and carry it as the
 642 C25 column of the same two tables. Its physics input is
 643 the unmodified regression deck that CLOUDY distributes
 644 for each case, `tsuite/auto/hii_paris.in` for HII40
 645 and `hii_coolstar.in` for HII20 — the same source,
 646 density, inner radius, and Lexington abundances as
 647 above — with only output commands appended; line lu-
 648 minosities are taken from a `save line list absolute`
 649 file and divided by the CLOUDY H β luminosity, and the

averaged temperature is CLOUDY’s own $n_e n_p$ -weighted volume average, the same weighting used on both other sides. The one geometric difference is that each deck runs to its own stopping criterion rather than to a prescribed outer radius — an electron fraction of 10^{-2} at HII40, the 1000 K temperature floor at HII20 — so it carries a little more of the partly neutral edge than a model truncated at R_{out} would. The radii reached, 1.467×10^{19} and 8.97×10^{18} cm, sit at the top of the published R_{out} ranges ($1.45\text{--}1.47 \times 10^{19}$ and $8.7\text{--}9.01 \times 10^{18}$ cm), so the two geometries are comparable, but the low-ionization rows fed by that edge — [O I], [S II], [N II] — are the ones where the difference could matter.

The temperature compared throughout this section is $\langle T[N_p N_e] \rangle$, the electron temperature averaged with the weight $n_{\text{HII}} n_e V$ — the same definition on both sides, evaluated leaf by leaf in MoCHII and cell by cell in the published models. The inner-edge temperature T_{inner} that the published tables also list is quoted separately where it is used. Ionic fractions are $n_e V$ -weighted means, the convention of MOCASSIN’s own output. As a further point of comparison we also ran MOCASSIN 2.02 itself, converged, on the HII40 model, and quote it below alongside the published columns.

Ionization structure. The helium and metal ionization is sensitive to how the ionizing band is discretized at the He I edge. On a plain logarithmic grid that edge is straddled and the He-ionizing photon budget is overdelivered (Section 3.1); pinning the bin edges on the thresholds returns that budget to the exact blackbody value and, with every other setting held fixed, moves the mean He I fraction from 0.058 to 0.163 and C III from 0.786 to 0.694, with O III and N III dropping correspondingly. It is the single largest lever on the ionization structure of this model, and the benchmark runs use the aligned band throughout. On it, the helium ratio $\langle \text{He}^+ \rangle / \langle \text{H}^+ \rangle$ is 0.715 at HII40, at the low end of the published 0.715–0.829 (median 0.767) and below a current CLOUDY (C25 0.741, itself under the published median), and 0.047 at HII20, on the published median of 0.048 inside a spread of 0.034–0.077.

Temperature. The weighted temperature is 7894 K at HII40, inside the published 7720–8199 K and 1.6% below the eight-code median of 8026 K, and 6376 K at HII20, 0.4% below the published floor of 6402 K and 4.3% below the median of 6662 K. The inner-edge temperature behaves the same way: 7270 K at HII40 against a published 7370–8318 K and 6752 K at HII20 inside a published 6562–7224 K. The converged MOCASSIN 2.02 run of the HII40 model gives 8176 K, also inside the published spread and 6% above MOCASSIN’s own

published entry. MoCHII therefore reproduces the published temperatures, and does so from the low side of the distribution in both cases.

That the two benchmarks sit at the same edge is informative, because the published columns are not themselves a stationary target. CLOUDY c25.00 on the same two models gives 8210 K and 6910 K, in both cases *above* the entire published spread; measured against a contemporary code rather than a 25-year-old table, MoCHII is 3.8% and 7.7% cooler. The two modern calculations thus straddle the published set from opposite sides, and the offset between them agrees in sign and size with the one documented approximation in the MoCHII thermal loop — the zero-density cooling function of Section 4.4, which overestimates the far-infrared line cooling at $n_e \sim 10^2 \text{ cm}^{-3}$ and so biases T_e low. The next subsection measures that overestimate directly.

5.3. Cooling-budget verification at a fixed state

An emission-line comparison cannot by itself separate an error in the atomic data from an error in the temperature, because each code’s temperature is the root of its own energy balance: too little cooling in one channel raises T_e , which raises every other channel, and the converged line ratios absorb part of the discrepancy. The cooling data can be compared free of that feedback only at a *fixed* state. CLOUDY c25.00 writes such a decomposition — its **save cooling** each output lists the cooling rate of every ion and every continuum channel zone by zone — so the test is to take its converged (T_e, n_e , ionic fractions) zone by zone, evaluate the MoCHII cooling terms on exactly that state, and integrate both over the model volume. What this measures is the absolute scale of two independent atomic-data compilations, channel by channel, with nothing else in either code entering. It is the comparison that isolates the cooling data from the temperature, which an equilibrium comparison cannot do, since the latter shows only the temperature at which each code’s own heating and cooling cross.

On the HII40 model the two agree within the accuracy of the underlying collision data. With the n -level emissivities evaluated at the local n_e , the ratio MoCHII/CLOUDY is 1.02 for [O III], 0.98 for [O II], 1.00 for [S III], 0.99 for [N II], 1.00 for [S II], 1.00 for [Ne II], 1.02 for [N III], 1.02 for [S IV], 0.99 for C III], 0.94 for [C II] and 0.99 for the H I collisional lines; summed over all line channels the two agree to 1.1%. Two channels fall outside $\pm 10\%$, [Ne III] (1.30) and [O I] (0.84), and they carry 1.9% and 0.1% of the budget. The free-free continuum, an independent check of the Gaunt factor and of the net ion charges entering it, agrees to 0.2%. Recombination cooling differs by bookkeeping rather than by data:

CLOUDY reports the free-bound loss net of the escaping recombination continuum, $4.02 \times 10^{37} \text{ erg s}^{-1}$, which falls between the MoCHII case-A and case-B sums (6.61 and 3.62×10^{37}), the difference being the ground-state term that the explicit diffuse field re-emits instead of losing. The heating side of the same CLOUDY model balances its cooling to 1.3×10^{-4} and deposits 4.57 eV for each ionizing photon, so the comparison state is itself an equilibrium one.

Replacing the n -level emissivities by the zero-density fit that the thermal loop uses (Eq. 9) raises the same summed line cooling to 1.116 times CLOUDY's. The excess is not spread over the ions: it is 5.1 percentage points of the line budget in [O III] and 3.9 in [C II], and in both it sits entirely in the far-infrared fine-structure lines, whose critical densities are at or below the local $n_e = 110 \text{ cm}^{-3}$. Since the line channels carry 79% of the cooling budget, that is a 9% excess in the total, and it accounts for the temperature bias quantitatively: scaling T_e on the fixed state until the MoCHII terms radiate the same total power — CLOUDY balances its own heating at the unscaled temperature, and the photoheating of the two codes agrees — gives 7784 K against CLOUDY's 8210 K, a 5.2% displacement, against the 3.8% by which the converged MoCHII run is cooler than C25. A first-order estimate at fixed ionization is thus more than sufficient to explain the offset, and the remedy is the $\Lambda(T, n_e)$ table of Section 4.4 rather than a change of atomic data.

Sensitivity to the other ingredients. The remaining modeling choices are smaller levers than the cooling function, and each is measured by an in-code experiment. The transport treatment is the largest of them: replacing the case-A explicit diffuse field by the case-B on-the-spot approximation, with every other setting fixed, gives 7570 K at HII40 and 6216 K at HII20, so the explicit field used here raises T_e by 4.3% and 2.6% and moves the two benchmarks the same way. Substituting MOCASSIN-style recombination (Péquignot 1991 radiative plus Nussbaumer–Storey 1983 dielectronic) for the Badnell 2023 data, whose $\alpha(\text{X III} \rightarrow \text{X II})$ are systematically lower (ratios 0.72–0.84), moves T_e by 0.2% and each metal ionization fraction by 0.026, so the recombination vintage is a minor lever on both. The dominant [O III] collision strengths are stable across database generations: the $^3\text{P}^{-1}\text{D}$ effective collision strength is 2.283 in the CHIANTI v7 compilation behind the published MOCASSIN column and 2.262 in v11 at 10^4 K , a 0.9% difference worth less than 30 K. The photoionization heating is standard, with mean photoelectron energies $\langle h\nu - 13.6 \rangle_{\text{H}} = 3.9 \text{ eV}$ and $\langle h\nu - 24.6 \rangle_{\text{He}} = 4.1 \text{ eV}$. Grid resolution is treated separately below.

Line ratios. The line intensities follow from that temperature and ionization structure rather than from the emission calculation, which is exact to the accuracy of its input data (Section 4.4: the n -level solver reproduces an independent reference solver to four digits, and the fitted collision data reproduce direct CHIANTI evaluations to 0.05–0.5%). At HII40 the infrared fine-structure lines, whose excitation energies are small enough that they are nearly temperature-independent, sit with the published set ([Ne III] $15.5 \mu\text{m}$ 0.278 in a published 0.263–0.429; [Ne II] $12.8 \mu\text{m}$ 0.202 in 0.177–0.217; [O III] $52+88 \mu\text{m}$ 2.41 in 2.10–2.42), and so do the temperature-sensitive optical lines of the same ion: [O III] 5007+4959 is 2.06 against a published 1.64–3.27, and the auroral [O III] 4363, the most temperature-sensitive entry in the tables, is 0.0033 against a published 0.0022–0.0070. The lines that fall outside the published ranges are examined below.

H β and the photon budget. A recombination line is a photon counter, and $L(\text{H}\beta)$ therefore tests whether the model is radiation-bounded: every ionizing photon absorbed by hydrogen returns as a recombination, and a fixed fraction of those recombinations produces H β . Both models here are bounded. At HII40 the ionization front is captured inside the prescribed sphere, at $1.43 \times 10^{19} \text{ cm}$ against an outer radius of $1.46 \times 10^{19} \text{ cm}$, and the outermost shell is 86% neutral; at HII20 the outer shells are fully neutral. The budget closes accordingly: summing the case-B recombinations over the converged solutions gives $0.953 Q_{\text{H}}$ and $0.971 Q_{\text{H}}$. What hydrogen does not absorb is taken up by helium, which at HII40 absorbs $0.114 Q_{\text{H}}$ and returns most of it to hydrogen through the He I diffuse channels. The predicted $L(\text{H}\beta) = 1.95 \times 10^{37}$ and $4.70 \times 10^{36} \text{ erg s}^{-1}$ are 5% and 4% below the published medians (3% below the lowest published values), matching the 4.7% and 2.9% of Q_{H} that the recombination budget assigns to helium rather than to hydrogen. The absolute H β luminosity is thus reproduced by photon counting, and adds no discrepancy of its own.

The complete published line list. Tables 2 and 3 repeat the comparison over the full benchmark line list in the form used by K. Wood et al. (2004), whose Tables 1 and 2 add their own Monte Carlo code (WME) to the eight other codes of the comparison; the MoCHII column is appended, and the C25 column runs beside its 1990s ancestor GF. Of the twenty lines tabulated at HII40, thirteen fall inside the published spread, one above it, and six below; of the sixteen at HII20, nine fall inside and seven below. The one line above the published range, [O I] 6300+6363 at HII40 (0.0158 against 0.0059–0.012), is fed by the thin partly neutral edge, where the pub-

Table 2. Lexington H II region benchmark HII40 (40 kK blackbody): the published multi-code results of *K. Wood et al. (2004, their Table 1)*, with a current CLOUDY c25.00 run and the MoCHII result appended. Line intensities are relative to $H\beta$.

Quantity	GF	C25	HN	DP	TK	PH	RS	RR	BE	WME	MoCHII
$L(H\beta)/10^{37} \text{ erg s}^{-1}$	2.06	2.06	2.02	2.02	2.10	2.05	2.07	2.05	2.02	2.01	1.95
He I 5876	0.119	0.100	0.112	0.113	0.116	0.118	0.116	...	0.114	0.114	0.108
C II] 2325+	0.157	0.189	0.141	0.139	0.110	0.166	0.096	0.178	0.148	0.172	0.154
C III] 1907+1909	0.071	0.062	0.076	0.069	0.091	0.060	0.066	0.074	0.041	0.078	0.029
[N II] 122 μm	0.027	0.030	0.037	0.034	...	0.032	0.035	0.030	0.036	0.031	0.032
[N II] 6584+6548	0.669	0.669	0.817	0.725	0.69	0.736	0.723	0.807	0.852	0.742	0.626
[N II] 5755	0.0050	0.0060	0.0054	0.0050	...	0.0064	0.0050	0.0068	0.0061	0.0057	0.0048
[N III] 57.3 μm	0.306	0.286	0.261	0.311	...	0.292	0.273	0.301	0.223	0.308	0.280
[O I] 6300+6363	0.0094	0.0112	0.0086	0.0088	0.012	0.0059	0.0070	...	0.0065	0.011	0.0158
[O II] 7320+7330	0.029	0.033	0.030	0.031	0.023	0.032	0.024	0.036	0.025	0.033	0.027
[O II] 3726+3729	1.94	2.32	2.17	2.12	1.6	2.19	1.88	2.26	1.92	2.23	1.90
[O III] 52+88 μm	2.35	2.40	2.10	2.26	2.17	2.34	2.29	2.34	2.28	2.42	2.41
[O III] 5007+4959	2.21	2.46	2.38	2.20	3.27	1.93	2.17	2.08	1.64	2.34	2.06
[O III] 4363	0.00235	0.0046	0.0046	0.0041	0.0070	0.0032	0.0040	0.0035	0.0022	0.0044	0.0033
[Ne II] 12.8 μm	0.177	0.200	0.195	0.192	...	0.181	0.217	0.196	0.212	0.192	0.202
[Ne III] 15.5 μm	0.294	0.216	0.264	0.270	0.35	0.429	0.350	0.417	0.267	0.263	0.278
[Ne III] 3869+3968	0.084	0.058	0.087	0.071	0.092	0.087	0.083	0.086	0.053	0.063	0.059
[S II] 6716+6731	0.137	0.304	0.166	0.153	0.315	0.155	0.133	0.130	0.141	0.18	0.299
[S II] 4068+4076	0.0093	0.0142	0.0090	0.0100	0.026	0.0070	0.005	0.0060	0.0060	0.0094	0.0123
[S III] 18.7 μm	0.627	0.504	0.750	0.726	0.535	0.556	0.567	0.580	0.574	0.623	0.494
[S III] 9532+9069	1.13	1.00	1.19	1.16	1.25	1.23	1.25	1.28	1.21	1.44	0.91
$10^3 \times \Delta(\text{BC } 3645)/\text{\AA}$	4.88	...	...	4.95	...	5.00	5.70	...	5.47	4.96	...
$T_{\text{inner}} / \text{K}$	7719	7812	7668	7663	8318	7440	7644	7399	7370	7743	7270
$\langle T[N_p N_e] \rangle / \text{K}$	7940	8210	7936	8082	8199	8030	8022	8060	7720	8183	7894
$R_{\text{out}}/10^{19} \text{ cm}$	1.46	1.47	1.46	1.46	1.45	1.46	1.47	1.46	1.46	1.45	1.43
$\langle \text{He}^+ \rangle / \langle \text{H}^+ \rangle$	0.787	0.741	0.727	0.754	0.77	0.764	0.804	0.829	0.715	0.771	0.715

NOTE—Codes, in the notation of *K. Wood et al. (2004)*: GF = Ferland’s CLOUDY; HN = Netzer’s ION; DP = Péquignot’s NEBU; TK = Kallman’s XSTAR; PH = Harrington’s code; RS = Sutherland’s MAPPINGS; RR = Rubin’s NEBULA; BE = Ercolano’s MOCASSIN; WME = the code of *K. Wood et al. (2004)* itself. Columns GF–BE agree entry by entry with Tables 4 and 5 of *B. Ercolano et al. (2003)*, the compilation of the published columns. C25 is CLOUDY c25.00 (*C. M. Gunasekera et al. 2025*) run for this paper, and is included because the published GF column dates from the 1990s version of the same code, so the two together measure how far that reference has moved. Its physics input is the unmodified c25.00 test-suite file `tsuite/auto/hii_paris.in` (HII40) or `hii_coolstar.in` (HII20) — CLOUDY’s own maintained version of these benchmarks — with only output commands appended; line luminosities come from a `save line list absolute` file and are divided by the CLOUDY H I 4861.32 Å luminosity, and the CLOUDY blends summed for each row are listed in `tables/make_wood_tables.py`. Unlike the published columns, C25 runs to the stopping criterion of its own deck rather than to a prescribed outer radius — an electron fraction of 10^{-2} at HII40, the 1000 K temperature floor at HII20 — which mainly affects the low-ionization rows fed by the partly neutral edge. The MoCHII column uses threshold-aligned 32 ionizing bins, the explicit Monte Carlo diffuse field including the He I channels (case A, matching the transfer treatment of the published codes), and Badnell 2023 recombination with the Mao & Kaastra 2016 ground-level split, at 4×10^7 stellar packets with the volume-integrated convergence criterion; the ionizing luminosity is set from the benchmark $L(\text{BB})$ so that $Q(\text{H})$ matches the specification; blends are the sum of the MoCHII components listed in `tables/make_wood_tables.py`, and MoCHII wavelengths are vacuum values. The MoCHII structural entries are computed from the same run: T_{inner} is the mean T_e of the cells at the illuminated inner edge, R_{out} the radius at which the shell-averaged x_{HII} falls through 0.5, and the helium ratio follows Eq. (17) of *B. Ercolano et al. (2003)*, $\langle \text{He}^+ \rangle / \langle \text{H}^+ \rangle = \int n_e n(\text{He}^+) dV / \int n_e n_p dV \times n_{\text{H}} / n_{\text{He}}$. Where the ionized gas still fills the outermost shell of the prescribed sphere, R_{out} would be set by the grid rather than predicted, and the entry is left empty. The Balmer jump is not computed. The $H\beta$ row is headed $10^{36} \text{ erg s}^{-1}$ in the published table; the values listed there are the $10^{37} \text{ erg s}^{-1}$ values of *B. Ercolano et al. (2003)*, and the heading is corrected accordingly here.

Table 3. Lexington H II region benchmark HII20 (20 kK blackbody): the published multi-code results of *K. Wood et al. (2004, their Table 2)*, with a current CLOUDY c25.00 run and the MoCHII result appended. Line intensities are relative to $H\beta$.

Quantity	GF	C25	HN	DP	TK	PH	RS	RR	BE	WME	MoCHII
$L(H\beta)/10^{36} \text{ erg s}^{-1}$	4.85	4.99	4.85	4.83	4.90	4.93	5.04	4.89	4.97	4.87	4.70
He I 5876	0.0072	0.0061	0.008	0.0073	0.008	0.0074	0.0110	...	0.0069	0.00684	0.0068
C II] 2325+	0.054	0.076	0.047	0.046	0.040	0.060	0.038	0.063	0.042	0.053	0.036
[N II] 122 μm	0.068	0.070	...	0.072	0.007	0.072	0.071	0.071	0.071	0.067	0.069
[N II] 6584+6548	0.745	0.809	0.786	0.785	0.925	0.843	0.803	0.915	0.846	0.778	0.624
[N II] 5755	0.0028	0.0032	0.0024	0.0023	0.0029	0.0033	0.0030	0.0033	0.0025	0.0025	0.0018
[N III] 57.3 μm	0.0040	0.0038	0.0030	0.0032	0.0047	0.0031	0.0020	0.0022	0.0019	0.0049	0.0036
[O I] 6300+6363	0.0080	0.0084	0.0060	0.0063	0.0059	0.0047	0.0050	...	0.0088	0.055	0.0092
[O II] 7320+7330	0.0087	0.0113	0.0085	0.0089	0.0037	0.0103	0.0080	0.0100	0.0064	0.0089	0.0064
[O II] 3726+3729	1.01	1.40	1.13	1.10	1.04	1.22	1.08	1.17	0.909	1.12	0.88
[O III] 52+88 μm	0.0030	0.0033	0.0026	0.0026	0.0040	0.0037	0.0020	0.0017	0.0022	0.0044	0.0037
[O III] 5007+4959	0.0021	0.0023	0.0016	0.0015	0.0024	0.0014	0.0010	0.0010	0.0011	0.0026	0.0020
[Ne II] 12.8 μm	0.264	0.296	0.267	0.276	0.27	0.271	0.286	0.290	0.295	0.289	0.289
[S II] 6716+6731	0.499	0.744	0.473	0.459	1.02	0.555	0.435	0.492	0.486	0.521	0.592
[S II] 4068+4076	0.022	0.023	0.017	0.020	0.052	0.017	0.012	0.015	0.013	0.017	0.015
[S III] 18.7 μm	0.445	0.209	0.460	0.441	0.34	0.365	0.398	0.374	0.371	0.381	0.204
[S III] 9532+9069	0.501	0.293	0.480	0.465	0.56	0.549	0.604	0.551	0.526	0.602	0.238
$10^3 \times \Delta(\text{BC } 3645)/\text{\AA}$	5.54	...	...	5.62	...	5.57	5.50	...	6.18	5.52	...
$T_{\text{inner}} / \text{K}$	7224	7155	6815	6789	6607	6742	6900	6708	6562	6852	6752
$\langle T[N_p N_e] \rangle / \text{K}$	6680	6910	6650	6626	6662	6749	6663	6679	6402	6692	6376
$R_{\text{out}}/10^{18} \text{ cm}$	8.89	8.97	8.88	8.88	8.7	8.95	9.01	8.92	8.89	8.73	8.69
$\langle \text{He}^+ \rangle / \langle \text{H}^+ \rangle$	0.048	0.043	0.051	0.049	0.048	0.044	0.077	0.034	0.041	0.047	0.047

NOTE—Codes, in the notation of *K. Wood et al. (2004)*: GF = Ferland’s CLOUDY; HN = Netzer’s ION; DP = Péquignot’s NEBU; TK = Kallman’s XSTAR; PH = Harrington’s code; RS = Sutherland’s MAPPINGS; RR = Rubin’s NEBULA; BE = Ercolano’s MOCASSIN; WME = the code of *K. Wood et al. (2004)* itself. Columns GF–BE agree entry by entry with Tables 4 and 5 of *B. Ercolano et al. (2003)*, the compilation of the published columns. C25 is CLOUDY c25.00 (*C. M. Gunasekera et al. 2025*) run for this paper, and is included because the published GF column dates from the 1990s version of the same code, so the two together measure how far that reference has moved. Its physics input is the unmodified c25.00 test-suite file `tsuite/auto/hii_paris.in` (HII40) or `hii_coolstar.in` (HII20) — CLOUDY’s own maintained version of these benchmarks — with only output commands appended; line luminosities come from a `save line list absolute` file and are divided by the CLOUDY H I 4861.32 Å luminosity, and the CLOUDY blends summed for each row are listed in `tables/make_wood_tables.py`. Unlike the published columns, C25 runs to the stopping criterion of its own deck rather than to a prescribed outer radius — an electron fraction of 10^{-2} at HII40, the 1000 K temperature floor at HII20 — which mainly affects the low-ionization rows fed by the partly neutral edge. The MoCHII column uses threshold-aligned 32 ionizing bins, the explicit Monte Carlo diffuse field including the He I channels (case A, matching the transfer treatment of the published codes), and Badnell 2023 recombination with the Mao & Kaastra 2016 ground-level split, at 4×10^7 stellar packets with the volume-integrated convergence criterion; the ionizing luminosity is set from the benchmark $L(\text{BB})$ so that $Q(\text{H})$ matches the specification; blends are the sum of the MoCHII components listed in `tables/make_wood_tables.py`, and MoCHII wavelengths are vacuum values. The MoCHII structural entries are computed from the same run: T_{inner} is the mean T_e of the cells at the illuminated inner edge, R_{out} the radius at which the shell-averaged x_{HII} falls through 0.5, and the helium ratio follows Eq. (17) of *B. Ercolano et al. (2003)*, $\langle \text{He}^+ \rangle / \langle \text{H}^+ \rangle = \int n_e n(\text{He}^+) dV / \int n_e n_p dV \times n_{\text{H}} / n_{\text{He}}$. Where the ionized gas still fills the outermost shell of the prescribed sphere, R_{out} would be set by the grid rather than predicted, and the entry is left empty. The Balmer jump is not computed.

lished entries themselves span a factor two and where the cell size of a three-dimensional grid is a real limitation. The rows below the range are largely the same in the two cases: He I 5876, [N II] 6584+6548 and 5755, and both [S III] entries.

The clearest reading of the tables is the contrast between the temperature-sensitive and the temperature-insensitive lines of the same ion, measured against C25 rather than against the 25-year-old columns. The far-infrared fine-structure lines, whose excitation energies are small enough that their emissivities barely depend on T_e , agree with C25 to a few percent at both benchmarks: [O III] 52+88 μm is $1.00\times$ and $1.12\times$ C25, [Ne II] 12.8 μm $1.01\times$ and $0.98\times$, [N III] 57.3 μm $0.98\times$ and $0.95\times$, and [S III] 18.7 μm $0.98\times$ at both. The temperature-sensitive optical and ultraviolet lines run low by about what the temperature offset implies: at HII40, where MOCHII is 3.8% cooler than C25, [O III] 5007+4959 is $0.84\times$ C25 and the auroral 4363 $0.72\times$; at HII20, where the offset is twice as large, [N II] 5755 is $0.56\times$, [O II] 7320+7330 $0.57\times$ and [O II] 3726+3729 $0.63\times$. At a given ionization structure, infrared lines flat and optical and auroral lines low is a temperature signature rather than an atomic-data one — which is what the fixed-state comparison of Section 5.3 establishes independently, since there the same emissivities agree with CLOUDY’s to 1% when both are evaluated on one state.

The C25 column also shows how much of the disagreement with the published columns is a property of those columns. [S II] 6716+6731 at HII40 is the clearest case: the GF entry is 0.137, C25 gives 0.304 and MOCHII 0.299, so what would be a $2.2\times$ excess against the 1990s column is a 2% difference against the current one, and both modern values sit inside the published spread only because one 2000-era entry (TK, 0.315) is high. Sulfur at HII20 is the same story in the other direction: [S III] 18.7 μm (0.204 against a published 0.34–0.46) and [S III] 9532+9069 (0.238 against 0.465–0.604) fall *below* every published code, and so does C25 (0.209 and 0.293), while [S II] 6716+6731 sits inside the published range for both (0.592 and 0.744). Both modern calculations therefore move the same S II/S III split in the same direction relative to the 2000-era compilation, which points at the sulfur data rather than at either code. A band experiment bounds one further difference: CLOUDY carries the full stellar spectrum, so photons below 13.6 eV ionize S^0 (10.36 eV) and C^0 (11.26 eV) in the partly neutral edge, while the benchmark band here starts at the hydrogen edge. Extending the MOCHII band down to 6 eV raises [S II] 6716+6731 by 6% at HII20 and by under 1% at HII40, with the other low-ionization lines moving by at most a few percent — a real ingredient of the

low-ionization rows, but far too small to account for the [S III] difference.

A second, structural difference is worth naming here, because it affects sulfur specifically. The published columns are not all built on the same set of ionization stages: the code of K. Wood et al. (2004), and CMA-CLONIZE after it, omit neutral sulfur and start the sulfur chain at S^+ , so that in gas where the ionizing field has been absorbed all sulfur is S^+ by construction; oxygen and neon in the same codes do carry their neutral stages. MOCASSIN instead solves the neutral stage of every element as an unknown, as MOCHII and CLOUDY do. The convention matters only outside the ionization front, where it fixes the S^+ reservoir that feeds [S II] rather than letting it recombine to S^0 ; it is one of the ingredients of the scatter among the published sulfur entries, and it is worth keeping in view when a single [S II] ratio is used to judge a code.

Radial structure. Figures 4 and 5 give the radial ionization and temperature structure in the panel layout of K. Wood et al. (2004), their Figures 1 and 2, so that the two can be read panel by panel. The morphology is the published one throughout. At HII40 the hydrogen and helium neutral fractions rise monotonically from 4×10^{-5} and 2×10^{-4} at the inner edge, the singly and doubly ionized fractions of carbon, nitrogen and oxygen cross within 0.15 pc of one another near 3.9 pc, C^{+3} stays below 0.05 and is confined to the inner region, and the temperature falls slightly out to about 2.2 pc before climbing steeply toward the front. At HII20 helium is essentially all neutral beyond 1.4 pc while hydrogen stays ionized out to 2.8 pc, so the doubly ionized metals occupy only a thin inner shell. Both models are radiation-bounded, so both outer radii are predictions. The ionized-volume-equivalent radius $(3V_{\text{ion}}/4\pi)^{1/3}$ and the shell-averaged $x_{\text{HII}} = 0.5$ crossing that the tables carry as R_{out} are 1.428 and 1.434×10^{19} cm at HII40 against a published 1.45–1.47 $\times 10^{19}$, and 8.58 and 8.69×10^{18} cm at HII20 against a published 8.7–9.01 $\times 10^{18}$: the two definitions of the outer boundary are not identical, but they locate the same front, and both sit 1–2% below the published ranges, which is what the recombination budget of 0.95–0.97 Q_{H} and the slightly larger α_B of a cooler nebula together imply. The temperature panels reproduce the published shape — a shallow inward decline followed by a steep rise at the front, where the surviving radiation field is hardest.

Grid resolution. The two three-dimensional codes of the comparison state what they used, and it is worth placing MOCHII beside them, because a Monte Carlo photoionization calculation on a Cartesian grid has a resolution as well as a photon budget. B. Ercolano et al.

(2003) ran the published MOCASSIN benchmark columns on $13 \times 13 \times 13$ cells with the source in a *corner*, so that the grid covers one octant of the spherically symmetric model; in their words, “although the number of sampling points along each orthogonal axis is only 13, this is the equivalent of a one-dimensional code with 273 radial points.” K. Wood et al. (2004) used 65^3 cells over the full box with the star at the center, which they describe as matching a 33^3 MOCASSIN octant, i.e. a cell width of “3% of the radius,” and 10^8 photon packets. The MoCHII runs here use a uniform level-6 octree, $64^3 = 262\,144$ leaves over the full box with the source at the center, and 4×10^7 packets. MoCHII and K. Wood et al. (2004) are therefore at essentially the same resolution — 4.6×10^{17} against 4.4×10^{17} cm on a side at HII40 — and both are about $2.5\times$ finer per axis than the octant grid behind the published MOCASSIN column. We did not repeat the benchmark on a deliberately coarsened grid, because the quantity at issue is already resolution-converged in the direction that matters: the front re-refinement test of Section 6 shows R_{eff} changing by $+0.008\%$ between a native level-7 grid and a coarser re-refined equivalent, and B. Ercolano et al. (2003) themselves attribute to grid coarseness only their inner-edge temperature T_{inner} — which their coarse cells make too *low*, since a cell carries the temperature of its center rather than of the illuminated face — and not $\langle T[N_{\text{p}}N_{\text{e}}] \rangle$ or the line ratios compared here. Grid resolution is thus not a candidate for the differences discussed above, with the exception of the thin partly neutral edge that feeds [O I] 6300+6363, which one cell width does not resolve in any of the three-dimensional codes.

How far the reference itself has moved. Because the C25 column is a re-run of one of the participating codes on the same two models, the difference between it and the GF column measures the drift of the reference over a quarter of a century, at fixed input. That drift is not small. The absolute $L(\text{H}\beta)$ is stable — 2.06 against 2.06 at HII40, 4.99 against 4.85 at HII20, in the units of the tables — as it must be, since it is fixed mostly by the photon budget. The temperature has risen by 270 K at HII40 and 230 K at HII20. The line ratios have moved much further: at HII40 [S II] 6716+6731 is $2.2\times$ its GF value, [O III] 4363 $2.0\times$, and [Ne III] 3869+3968 $0.69\times$; at HII20 the two [S III] entries fall to 0.47 and 0.58 of their GF values while [S II] 6716+6731 rises by $1.5\times$. A benchmark column published in 2000 is therefore not a fixed target for a code written in 2026, and a modern code that reproduced the GF entries exactly would be reproducing superseded atomic data. This is the reason the comparison above is made against the published

spread and against C25 rather than against any single published column.

Where MoCHII sits among the codes. D. Péquignot et al. (2001) and B. Ercolano et al. (2003) note that the eight codes do not scatter randomly: GF, HN, and DP agree more tightly with one another — and, in the planetary-nebula cases, with MOCASSIN — than with TK, PH, and RS, a coherence they attribute to more recently updated atomic data and to a more careful treatment of the diffuse radiation field. A second grouping runs along the transfer treatment: the codes that solve the transfer exactly (RR, PH, and MOCASSIN) give systematically lower temperatures at the illuminated inner edge, because the radiation field there is softer than an approximate treatment makes it. MoCHII belongs with the first group on the data axis — exact hydrogenic and Verner cross sections, Badnell 2023 recombination, CHIANTI v11 collision strengths, the same vintage CLOUDY uses — and with the second on the transfer axis, since the analytic direct-field estimator and the explicit diffuse field used for these benchmarks are an exact treatment in three dimensions; its inner-edge temperature is correspondingly at or below the low end of the published T_{inner} range. Its ionization structure, its outer radius, its $\text{H}\beta$ luminosity and most of its line ratios land inside the published spread, with helium at the low edge of it in the same direction as a current CLOUDY, and its weighted temperature is inside the spread at HII40 and 0.4% under it at HII20. What remains is a coherent and quantified bias rather than an open discrepancy: MoCHII runs 4–8% cooler than a contemporary CLOUDY on both models, in the direction and by about the amount that the zero-density cooling function of Section 4.4 implies, and the density-dependent $\Lambda(T, n_{\text{e}})$ that removes it is the next item for the thermal balance.

6. SOLUTION-DRIVEN FRONT REFINEMENT

A distinguishing capability of the octree grid is that the refinement can be driven by the ionization solution itself. At a chosen iteration the octree is rebuilt: a leaf is flagged as “front” when $\epsilon < x_{\text{HI}} < 1 - \epsilon$ or a face neighbor differs by more than a threshold; the new tree carries the maximum level wherever a cell overlaps a front-flagged leaf and a base level elsewhere. The gas state and opacity map over by position, and every dependent array is recreated. All ranks rebuild identically from the shared state, and the old grid’s shared-memory windows are freed immediately after the position sampling, so repeated re-refinement neither leaks nor holds two grids in memory at once.

On the Strömgren configuration, starting at level 5 and rebuilding to level 7 at iteration 15 ($32\,768 \rightarrow$

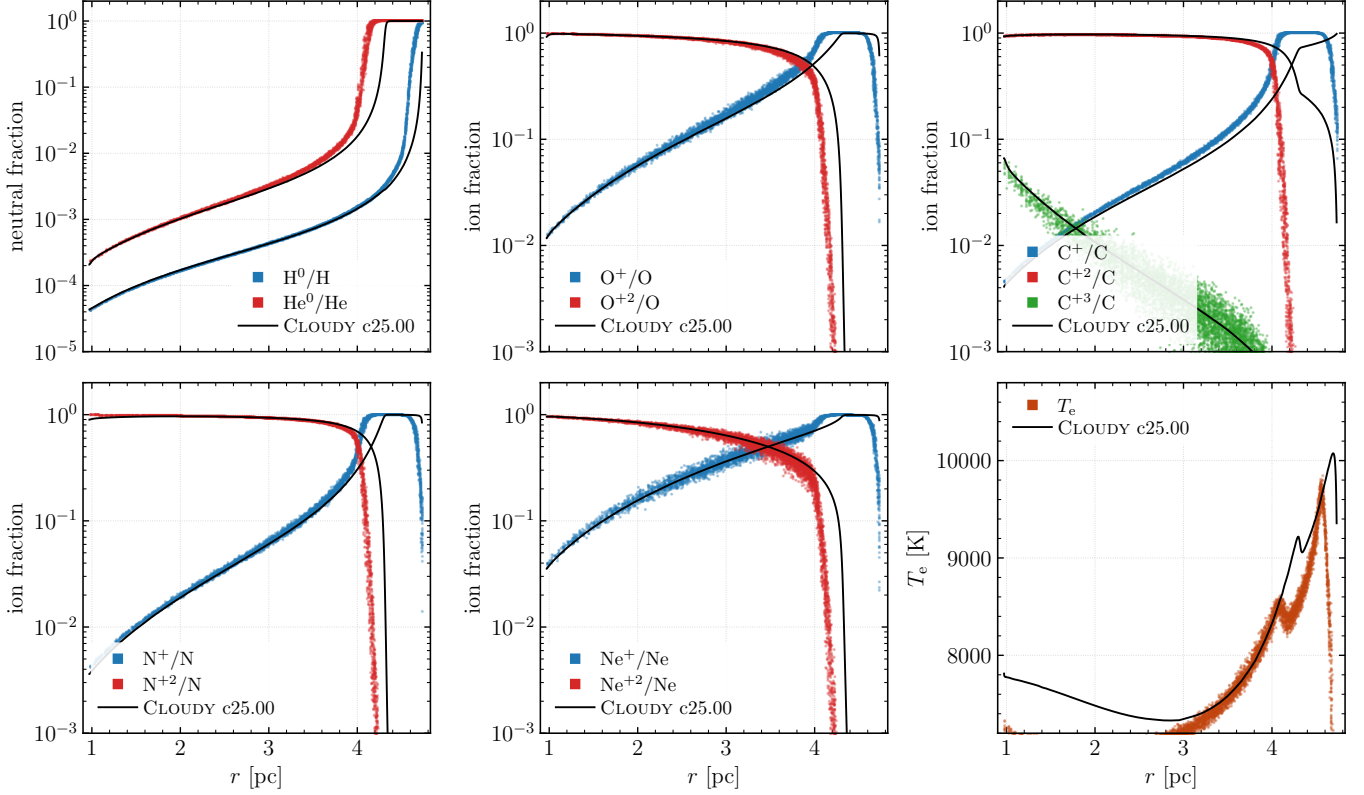


Figure 4. Lexington HII40 benchmark: radial ionization and temperature structure of the MoCHII solution, in the panel layout of K. Wood et al. (2004), their Figure 1. Each point is one octree leaf inside the prescribed shell, thinned at random for legibility; the temperature panel shows the leaves that are less than 95% neutral, the convention of K. Wood et al. (2004). Nitrogen stops at N^{+2} because the registry carries three nitrogen stages for this model. This is the same run as the MoCHII column of Table 2, on the threshold-aligned 32-bin ionizing grid. Black lines are CLOUDY c25.00 on the same model — the run behind the C25 column of Table 2, whose input is the unmodified c25.00 test-suite deck `tsuite/auto/hii_paris.in` — drawn as a line as K. Wood et al. (2004) draw CLOUDY in their own figures. The CLOUDY profile is cut at the outer radius of the MoCHII shell so that both codes cover one volume, and the 95% cut is applied to its zones as well. In the temperature panel the MoCHII points run below the CLOUDY curve through the interior and cross it at the front.

464 311 leaves), the re-refined grid gives $R_{\text{eff}} = 3.0432$ pc
 465 against a native uniform level-7 grid's 3.0430 pc
 466 (2 097 152 leaves) — agreement to +0.008% with $4.5\times$
 467 fewer cells, and both improve on level 6 (+0.56%), as
 468 resolution convergence expects (Figure 6). The front
 469 profiles overlies exactly.

As a demonstration on a realistic geometry, MoCHII
 was run on an Illustris-TNG 60-kpc cutout (density
 and neutral-fraction columns from the simulation), with
 a $Q_{\text{H}} = 10^{53} \text{ s}^{-1}$, 40-kK source at the density peak
 and thermal balance, metals, and re-refinement enabled
 (Figure 7). Dense clumps cast sharp neutral shadows,
 the circumgalactic gas sits at $x_{\text{HI}} \sim 10^{-4}$ with an n_e -
 weighted mean $T_e = 7502$ K, and the refinement (32 768
 $\rightarrow$ 617 324 leaves) tracks the *distributed* ionization topol-
 ogy of a turbulent medium — clump skins and partially
 ionized gas rather than a single thin shell. This is the
 regime for which the volume-integrated convergence cri-
 terion (Section 3.5) was designed.

7. DUST AND PAH COUPLING

Because MoCHII is built on a dust radiative-transfer engine, the dust physics is a first-class part of the transport. The coupling has four parts.

Absorption competition. Each band bin adds a dust absorption opacity $\kappa_d = \rho \kappa s_{\text{abs}}(b)$ with $s_{\text{abs}}(b) = (1 - a(E_b)) C_{\text{ext}}(E_b) / C_{\text{ext}}(\lambda_{\text{ref}})$ from a WD01/D03 $R_V = 3.1$ extinction table (J. C. Weingartner & B. T. Draine 2001; B. T. Draine 2003) covering the EUV, where $C_{\text{abs}}/\text{H} = 1.83 \times 10^{-21}$ at 13.6 eV falls to $3.96 \times 10^{-22} \text{ cm}^2$ at 100 eV. On the dusty Strömgren test, against a 1D reference carrying identical dust absorption, the full MW dust gives $R_{\text{eff}} = 2.246$ pc versus the reference 2.231 pc (+0.69%), reproducing the classic dusty reduction $R/R_0 = 0.735$ (Figure 8); dust absorbs 59% of the band luminosity there.

Scattering. With scattering enabled the scattering opacity joins the extinction (in-band Henyey–Greenstein asymmetry $g = 0.67\text{--}0.99$, strongly forward), and inter-

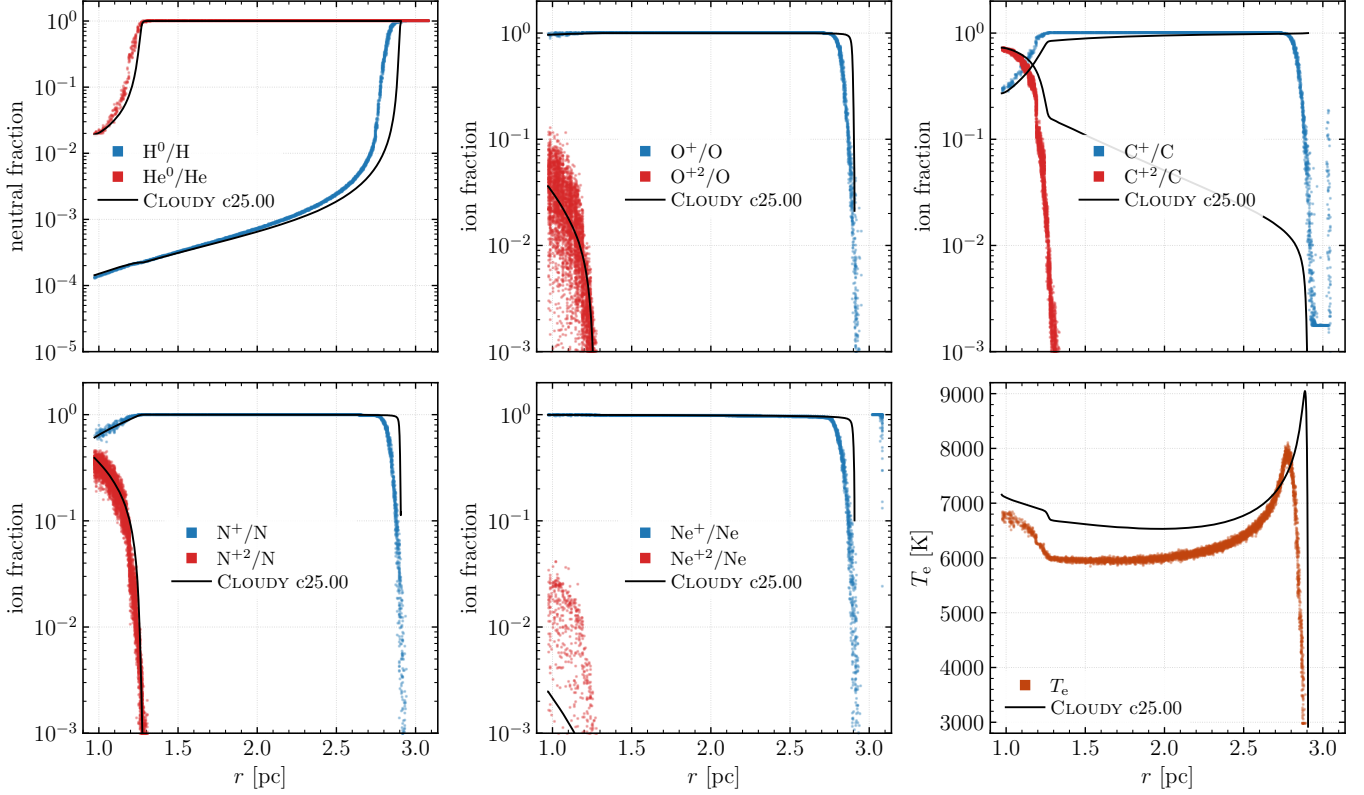


Figure 5. Lexington HII20 benchmark, in the panel layout of K. Wood et al. (2004), their Figure 2; conventions as in Figure 4. Ne^{+2} is confined within 1.4 pc, and C^{+3} stays below the plotted range and carries no series. This is the same run as the MoCHII column of Table 3. At 20 kK the helium-ionizing photons are so few that helium is neutral beyond 1.4 pc, which is why the aligned ionizing grid matters much less here than at HII40. Black lines are CLOUDY c25.00 on the same model, from the unmodified c25.00 deck `tsuite/auto/hii_coolstar.in` behind the C25 column of Table 3. CLOUDY stops on its own criterion at 2.91 pc, short of the outer radius of the MoCHII shell, so its curves end there, and its final near-vertical fall is the recombination front itself, resolved in zones far finer than a MoCHII cell.

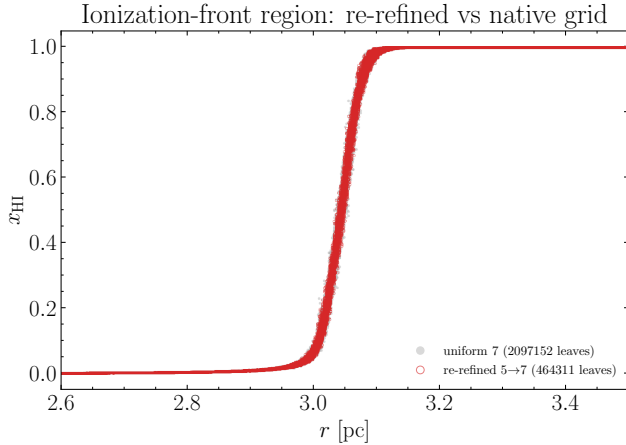


Figure 6. Re-refinement test: the front profiles of the re-refined 5 → 7 octree and the native uniform level-7 grid overlaid exactly, at $4.5\times$ fewer cells.

Strömgren radius, 2.237 pc, lands inside the analytic bracket (2.086 pc if the dust extinction is all absorption, 2.246 pc for absorption only) and close to the absorption-only bound, exactly as strong forward scattering predicts.

Live ionization-dependent dust with a PAH split. Following P. Laursen et al. (2009), the dust content is traced by the metallicity-scaled hydrogen, with neutral gas carrying the full dust column and ionized gas only a survival fraction f_{ion} of it (grains are eroded where the hydrogen is ionized). MoCHII evaluates this on the *computed* neutral fraction and refreshes it every iteration, and it splits the grains into a large-grain share $1 - f_{PAH}$ and a PAH share f_{PAH} with independent ionized-gas survival, so the dust opacity of a leaf is

$$\kappa_d \propto \frac{Z}{Z_{ref}} n_H [(1 - f_{PAH})(x_{HI} + f_{ion} x_{HII}) + f_{PAH}(x_{HI} + f_{ion,PAH} x_{HII})], \quad (10)$$

with $x_{HII} = 1 - x_{HI}$ and $f_{ion,PAH} = 0$ by default (PAHs destroyed in ionized gas). Because this dust opacity

actions are sampled by a forced-first-flight scheme after the analytic direct walk, with an albedo weight at each interaction and Russian roulette. The full-scattering

MoCHII on an Illustris-TNG cutout: $Q_H = 10^{53} \text{ s}^{-1}$ source, $z = 4.7$ kpc plane

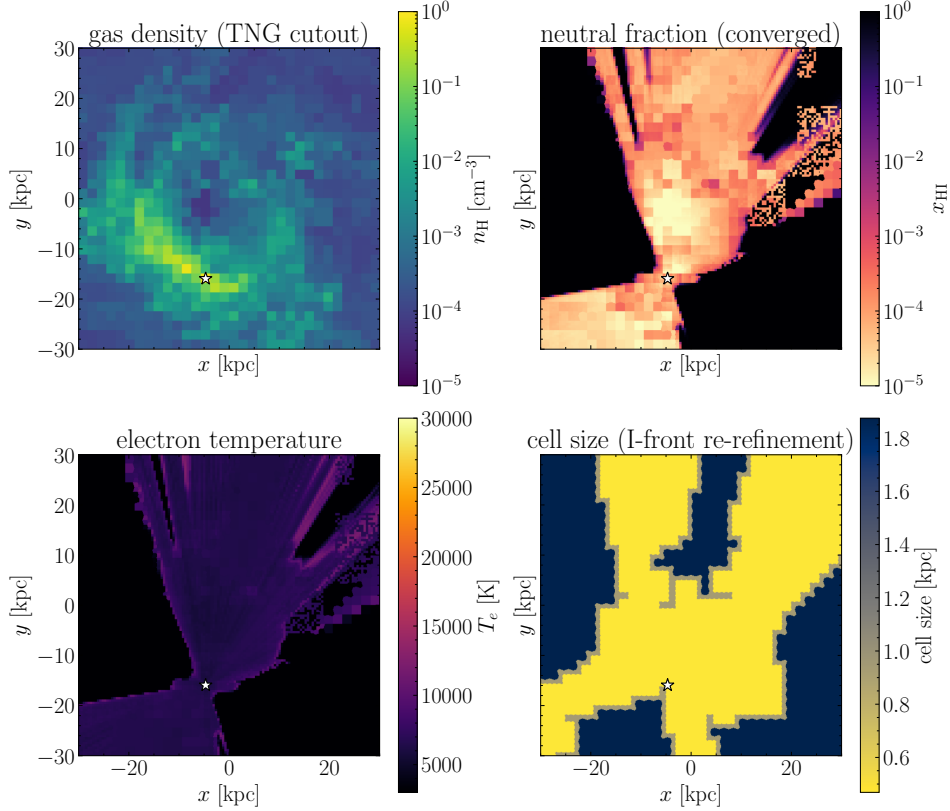


Figure 7. MoCHII on an Illustris-TNG cutout: gas density, converged neutral fraction (note the clump shadows), electron temperature, and the re-refined cell size, in the source plane.

is rebuilt from the ionization state each iteration while that state is in turn set by the dust-attenuated field, the dust-ionization feedback loop is solved self-consistently; it converges to a near dust-free interior ($R/R_0 = 0.996$).

Figure 8 contrasts the three treatments on the same Strömgren configuration. Static full MW dust competes with hydrogen for the ionizing photons throughout the volume, so the ionization front is pushed inward to $R_{\text{eff}} = 2.25$ pc, the classic dust-limited size ($R/R_0 = 0.736$). Under `laursen09_live`, by contrast, the interior dust vanishes together with the neutral fraction it is tied to — once the gas is ionized it carries essentially no dust to absorb the ionizing field — so the front profile overlies the dust-free case almost exactly ($R_{\text{eff}} = 3.04$ pc, $R/R_0 = 0.996$). The physical distinction is that only the neutral gas beyond the front and its thin skin retain dust, which is where the dust column that matters for the escaping radiation actually sits.

Grain heating, equilibrium temperature, and IR emission. The transported grain heating drives the equilibrium dust temperature by balancing the absorbed power against $4\pi \int s_{\text{abs}}(\nu) B_\nu(T) d\nu$; with the band widened to 6 eV, T_d runs from 89 K at 0.3 pc to 28 K at the

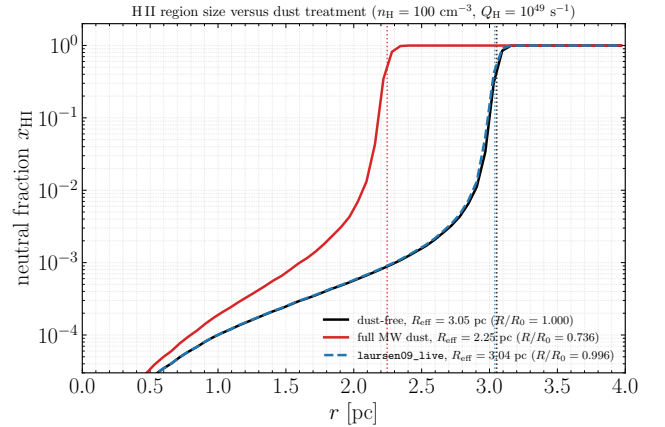


Figure 8. H II region size versus dust treatment: the radial neutral-fraction profile $x_{\text{HI}}(r)$ for the dust-free, full MW dust, and `laursen09_live` runs of the same Strömgren configuration ($n_H = 100 \text{ cm}^{-3}$, $Q_H = 10^{49} \text{ s}^{-1}$, $T_\star = 4 \times 10^4$ K, case B). Dotted vertical lines mark each effective ionized radius R_{eff} . Full MW dust pushes the ionization front inward to $R/R_0 = 0.736$, while the live ionization-dependent dust vanishes in the ionized interior and recovers the dust-free size ($R/R_0 = 0.996$).

1134 sphere edge (the canonical H II-region gradient), and the

IR spectrum integral closes on the absorbed power to $L_{\text{IR}}/L_{\text{abs}} = 1.000$. A full stochastic dust emission solve (the MoCAFE SEDust module; astrodrust, DL07, and Zubko grain models with PAH bands, after B. T. Draine & A. Li 2007) is available at output time: it uses the size-resolved stochastic-heating spectrum as a *shape* normalized to the transported absorbed power, so the normalization is exact by construction. With the live PAH share (Figure 9), the PAH bands drop to 0.15–0.26 of the PAH-everywhere reference and the projected $8\mu\text{m}$ map turns from centrally peaked into a *limb-brightened annulus* just inside the ionization front — the classic Spitzer/JWST bubble morphology.

8. THE PHOTODISSOCIATION-REGION SKIN

With the FUV band extension and three optional switches (all off by default), MoCHII models the atomic photodissociation-region (PDR) skin just outside the ionization front. The switches feed the metal-cascade electrons into the n_e closure, add metal photoheating to the thermal balance, and add grain photoelectric heating and grain recombination cooling with the E. L. O. Bakes & A. G. G. M. Tielens (1994) fits driven by the local FUV field G_0 , together with the H-impact fine-structure cooling of [C II] $158\mu\text{m}$ and [O I] $63\mu\text{m}$ (Ġ. Barinovs et al. 2005). The last is essential: the standard cooling fits are electron-impact only, and with $n_{\text{HI}}/n_e \sim 10^3\text{--}10^4$ in the PDR the hydrogen collisions are the coolant. Two guards keep the photoelectric heating physical (evaluated at $\min(T, 10^4\text{ K})$ so the fits stay in range, and weighted by x_{HI} so ionized gas, whose grains are EUV-charged and whose energy is already routed to the IR, is not overheated). Both are necessary: without them the fits are evaluated outside their validity range and the zone runs away to a false 12–35 kK.

Figure 11 shows the converged radial structure of a PDR test (a $n_{\text{H}} = 100$ octree sphere with a central source and all three switches on). The interior is fully ionized with a sharp front; beyond the front, the C II fraction is unity, the electron density falls to the carbon-dominated value $n_e \simeq A_C n_{\text{H}}$, and the electron temperature settles to a genuine PDR equilibrium (a $\sim 336\text{ K}$ to $\sim 211\text{ K}$ gradient, not pinned at the temperature floor). This is the atomic PDR skin only: there is no H_2/CO chemistry or line transfer, so the PDR temperatures are first-order estimates, adequate for the ion structure and rough [C II] emissivities.

9. SUMMARY AND OUTLOOK

MoCHII is a three-dimensional Monte Carlo photoionization code for dusty nebulae on adaptive octree grids, built by adding a self-consistent gas-physics layer

to a validated dust radiative-transfer engine. Its design choices — octree AMR with solution-driven front refinement, a zero-variance analytic direct-field estimator with a dedicated Cartesian DDA backend, an explicit optional Monte Carlo diffuse field, a data-driven trace-metal registry with modern atomic data, first-class dust absorption/scattering and equilibrium/stochastic emission with PAH physics, and peel-off synthetic imaging — distinguish it from the uniform-grid and 1D codes. It reproduces analytic attenuation to +0.004%, the Strömgren sphere to +0.56%, and a thermal H/He nebula to 0.14%. On the Lexington 2000 H II region benchmarks, with the ionizing bins aligned on the ionization thresholds — the discretization that controls the helium and metal ionization structure — it lies inside the published multi-code spread: the $n_{\text{p}}n_e$ -weighted temperature is 7894 K at HII40 within a published 7720–8199 K and 6376 K at HII20 just under a published floor of 6402 K; both models are radiation-bounded, so both outer radii are predicted, and $L(\text{H}\beta)$ follows the recombination budget to a few percent. The line-cooling data are checked independently of the temperature by evaluating the MoCHII cooling terms on a fixed state taken from a CLOUDY c25.00 cooling decomposition: channel by channel the two compilations agree, and the summed line cooling agrees to 1%. The one documented approximation left in the thermal loop is the zero-density cooling function, which through the far-infrared fine-structure lines overestimates the total line cooling by 12% at $n_e = 110\text{ cm}^{-3}$ and places both benchmarks at the cool edge of the published range, 4–8% below a CLOUDY c25.00 run of the same two models carried as an extra column of both tables; a density-dependent $\Lambda(T, n_e)$ is the planned replacement. That comparison column also shows how far the reference has moved in a quarter of a century: c25.00 sits above every published entry in temperature, and it moves the same S II/S III rows in the same direction MoCHII does. The solution-driven re-refinement reproduces a native high-resolution grid to +0.008% with $4.5\times$ fewer cells, the dust coupling closes energy exactly and reproduces the limb-brightened Spitzer/JWST PAH morphology, and the atomic PDR skin recovers the carbon-dominated electron density and a physical PDR temperature.

Development is ongoing along several lines. A plane-parallel (xy-periodic slab) illumination mode with an emergent $I(\mu)$ boundary tally extends the code to slab and grazing-illumination geometries. A He I 2^3S metastable diagnostic supports synthetic 10830 Å absorption modeling. A hard-spectrum extension for planetary nebulae and active galactic nuclei — higher photon energies, extended ion stages, inner-shell/Auger

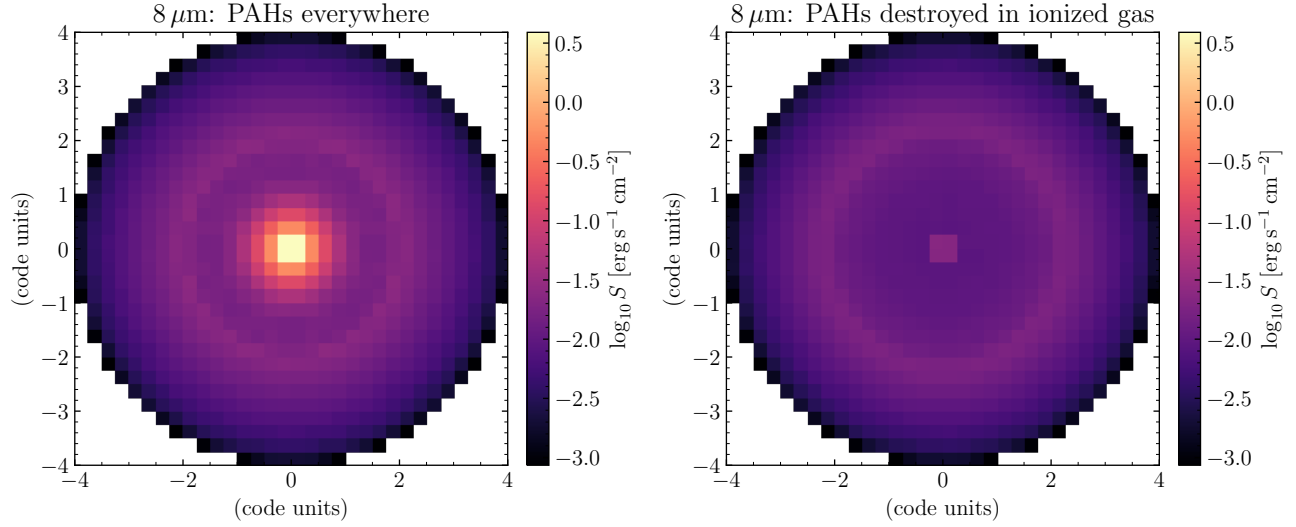


Figure 9. Stochastic dust emission at $8\,\mu\text{m}$ with the ionization-dependent PAH share. Both panels are two-dimensional projection maps — the emissivity integrated along the line of sight into a surface brightness — and share one brightness scale. Left: PAHs surviving everywhere, giving a centrally peaked image. Right: PAHs destroyed in the ionized interior (x_{HII} -weighted survival). The central peak disappears and the projected brightness instead rises outward to a broad limb-brightened annulus just inside the ionization front, a factor 1.9 above the depressed interior.

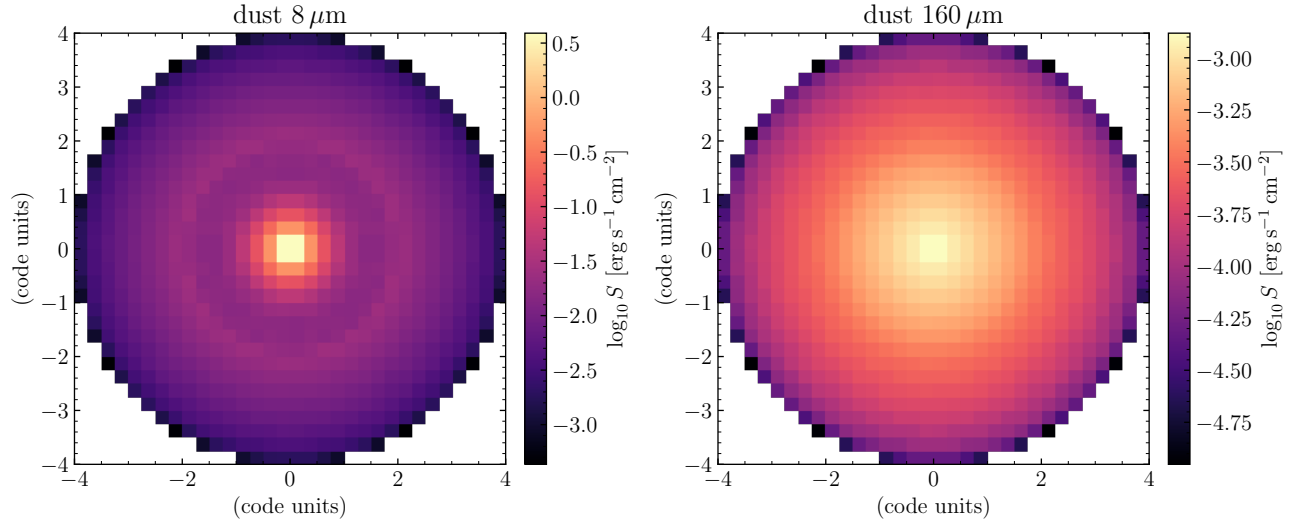


Figure 10. Dust emission at 8 and $160\,\mu\text{m}$ from the stochastic SEDust solve, as two-dimensional projection maps of the surface brightness. The short wavelength is centrally peaked (warm inner dust) while the far-infrared is spatially broader, the projected signature of the dust temperature gradient.

processes, and density-dependent cooling tables — is planned. The trace-metal registry makes further ions a data operation as the science requires them. Together these make MoCHII a tool for modern three-dimensional synthetic observations of H II regions, planetary nebulae, and the illuminated interstellar and circumgalactic medium.

ACKNOWLEDGMENTS

Software: MoCHII (<https://github.com/seoncafe/MoCHII>), MoCafe (<https://github.com/seoncafe/MoCafe>), CHIANTI (K. P. Dere et al. 1997; R. P. Dufresne et al. 2024), MOCASSIN (B. Ercolano et al. 2003), Cloudy (M. Chatzikos et al. 2023), astropy, matplotlib, NumPy

APPENDIX

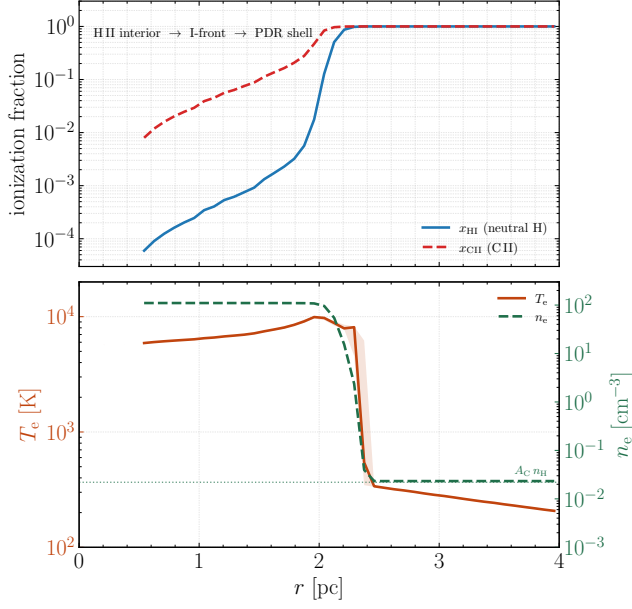
Photodissociation-region model: radial profiles ($n_{\text{H}} = 100 \text{ cm}^{-3}$ sphere)


Figure 11. Radial structure of the PDR test: the neutral fraction, electron temperature, and electron density as a function of radius. Inside the front the gas is fully ionized; beyond it the electron density drops to the carbon-dominated value and the temperature settles to a PDR equilibrium above the floor.

A. FITTED ATOMIC-DATA COEFFICIENTS

The atomic data are fitted offline and evaluated online (Section 4.4). Table 4 gives the quality of the line-cooling fits of every ion and Table 5 summarizes the n -level diagnostic models; it is generated mechanically from the data files by `paper/tables/make_atomic_tables.py`. The coefficients themselves — the amplitudes and excitation temperatures of the line-cooling fits, and the Burgess–Tully Chebyshev $\Upsilon(T)$ sets of the n -level models, several for each transition — are distributed with the code rather than reproduced here.⁵

⁵ The fitted line-cooling coefficients of every ion are provided as text in `data/atomic/cooling_fit_parameters.txt` (and, one file for each ion, in `data/atomic/cooling_<ion>.txt`), with the fit form, units, fit range, and maximum fit error documented in `data/atomic/README.md`; the n -level models are in `data/atomic/nlevel_<ion>.txt`. Every file carries a provenance header.

Table 4. Quality of the collisional line-cooling fits: the number of exponential terms N , the maximum fit error of each ion over the full 10^3 – 10^5 K range, and the maximum error restricted to the temperatures that carry the cooling ($\Lambda > 10^{-3}\Lambda_{\text{max}}$). The two agree for most ions; where they differ, the large full-range error sits in a temperature tail in which the ion radiates negligibly. The fitted coefficients themselves are distributed with the code (Appendix A).

Ion	N	max err	where it cools	Ion	N	max err	where it cools
H I	4	0.52%	0.52%	Si IV	3	0.61%	0.61%
C I	6	0.27%	0.27%	Ar III	2	3.90%	3.90%
C II	5	0.35%	0.35%	Ar IV	4	0.27%	0.27%
C III	6	5.48%	0.50%	Mg II	4	0.22%	0.12%
N I	5	0.58%	0.34%	Fe II	5	0.56%	0.56%
N II	5	0.66%	0.66%	Fe III	5	0.48%	0.48%
N III	5	0.70%	0.70%	Si II	6	0.91%	0.91%
O I	6	0.20%	0.20%	Si III	6	27.70%	11.29%
O II	4	0.27%	0.15%	Si IV	2	0.90%	0.90%
O III	5	0.32%	0.32%	Cl II	3	8.05%	8.05%
Ne II	5	0.40%	0.40%	Cl III	6	9.15%	9.15%
Ne III	5	0.34%	0.34%	Cl IV	2	7.30%	7.30%
S II	3	2.23%	2.16%	Ca II	4	1.03%	0.42%
S III	5	0.23%	0.23%	Ca V	2	4.34%	4.34%

Table 5. n -level diagnostic models: number of levels, number of fitted effective-collision-strength transitions, and the worst transition fit error. The level energies and radiative A -values are from CHIANTI v11.0.2, and $\Upsilon(T)$ is a low-order Chebyshev fit in Burgess–Tully scaled space over 10^3 – 10^5 K; the full coefficient sets are in the provenance-headed files `data/atomic/nlevel_<ion>.txt`.

Ion	levels	transitions	worst err	Ion	levels	transitions	worst err
H I	25	45	0.20%	Ar III	5	10	0.19%
C II	5	10	0.56%	Ar IV	5	10	0.20%
C III	5	10	0.35%	Mg II	3	3	0.15%
N II	5	10	0.17%	Fe II	13	78	0.20%
N III	5	10	0.45%	Fe III	14	91	0.20%
O I	5	10	0.17%	Si II	5	10	0.19%
O II	5	10	0.27%	Si III	5	10	0.65%
O III	6	15	1.68%	Si IV	3	3	0.54%
Ne II	2	1	0.16%	Cl II	5	10	0.17%
Ne III	5	10	0.19%	Cl III	5	10	0.19%
S II	5	10	1.25%	Cl IV	5	10	0.19%
S III	6	15	0.29%	Ca II	5	10	0.19%
S IV	5	10	0.18%	Ca V	5	10	0.19%

NOTE—Expanded 16/34-level Fe II/III models (`nlevel_fe_{2,3}_full.txt`) are available for iron-line studies. Ar II, Cl V, and Ca III/IV have no CHIANTI v11 level data and are tracked in the ionization balance without lines.

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